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Smart textiles for XR: a design space based on state-of-the-art research

Mengqi Jiang · Tianchen Dai · Ruoqing Yan · Rongkai Shi · Ziqian Bai · Qianwen Tan · Jun Huang · Hai-Ning Liang

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Abstract
Smart textiles, integrating electronic components into fabrics, enhance Extended Reality (XR) interactions by enabling intuitive, haptic, and embodied experiences. This paper addresses three key questions about smart textiles in XR—their current role, design space, and user perceptions—via a scoping review of 109 publications, a novel design framework, case studies, and an empirical user study. The review synthesizes trends, such as the dominance of on-body forms like gloves in Virtual Reality (VR) for haptic feedback, while identifying gaps in Augmented Reality (AR)/Mixed Reality (MR) and near/off-body textiles. Further, we proposed a design space structured around Textile Form, Interaction Mode, and Application Domain to map synergies and opportunities, demonstrated via two cases: a smart pillow for haptic social interactions in VR and wearable keyboard sleeves for text entry in MR. Besides, an exploratory image-based perception study suggests that familiar on-body forms and gesture/tactile interaction concepts were perceived relatively favorably, while some biodata-related concepts raised anticipated comfort and privacy concerns. These results provide framework and insights for HCI researchers and designers, fostering innovation in textile-augmented XR that bridges physical and digital realms.

Keywords Design space · Virtual reality · Extended reality · Smart textile · HCI

1 Introduction

Smart textiles, also known as e-textiles, integrate electronic components, sensors, and actuators into flexible textile structures, enabling them to sense, respond, and adapt to environmental or user stimuli (Weng et al. 2016; Jiang et al. 2024). Emerging from early experiments in wearable computing since the 1990s (Mann 1997), smart textiles have evolved from niche prototypes to practical applications in fields such as healthcare (Libanori et al. 2022), fashion (Uğur 2013), and materials science (Islam et al. 2020). This interdisciplinary domain leverages advances in nanotechnology, conductive fibers, and embedded computing to create seamless, unobtrusive interfaces that blend with everyday life.

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Ruoqing Yan, Rongkai Shi, Ziqian Bai, Qianwen Tan, Jun Huang have contributed equally to this work.

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In parallel, Extended Reality (XR)—encompassing Virtual Reality (VR), Augmented Reality (AR), and Mixed Reality (MR)—has transformed human-computer interaction (HCI) by merging digital and physical worlds (Cortes et al. 2024). VR immerses users in fully simulated environments, while AR/MR overlays virtual elements onto reality, enabling applications in gaming, training, education (Samala et al. 2024), etc. However, traditional XR systems often prioritize visual and auditory feedback, leading to limitations in embodiment and realism due to the absence of rich haptic and tactile sensations (Ko and Rogers 2021). Recent advancements highlight the need for multisensory integration to enhance immersion, reduce cybersickness, and support natural interactions (Karunanayaka et al. 2021).

The convergence of smart textiles and XR holds immense promise for addressing these gaps, creating embodied experiences that are engaging (Tomico and Wilde 2016), practical (Shi et al. 2020), and intuitive (Ruckdashel et al. 2021). For instance, textile-based haptics can simulate touch in remote social VR scenarios or provide biofeedback in AR training simulations, improving user experience, facilitating data acquisition, and unlocking novel domains like rehabilitation and remote collaboration (Tran et al. 2023; Yoon et al. 2024; Liu et al. 2021). Despite these benefits, research remains fragmented, with most studies focusing on isolated aspects of either smart textiles or XR applications. There remains a notable gap in research that integrates these two domains to uncover their synergistic potential—particularly in bridging physical and digital interactions. For instance, envision embracing a smart pillow that conveys haptic 'hugs' across virtual distances in VR. To address this, we propose three research questions (RQs):

- RQ1: How do smart textiles function in current XR interaction research?
- RQ2: What is the design space for the XR-textile interaction, including the textile form and the interaction mode?
- RQ3: How do users perceive the usefulness, anticipated real-world use, and expected comfort of representative XR-textile interaction concepts derived from the design space?

To address RQ1, we conducted a scoping review of the literature on smart textiles in XR applications. Following the PRISMA guidelines (Tricco et al. 2018), we systematically searched academic databases using keyword sets related to textiles and XR technologies. This yielded 2,881 initial records, refined through rigorous screening to a final corpus of 109 relevant publications. The review categorizes findings into **corpus characteristics**, **textile form**, **interaction mode**, and **application scenarios**, see Sect. 2.

For RQ2, we synthesized a design space framework from the scoping review result, see Sect. 3. This framework maps existing synergies and underexplored gaps. Further, we illustrate the design space with examples from the literature and two case studies: a smart pillow for VR social interaction providing haptic "hugs" and wearable keyboard sleeves for MR typing, demonstrating practical applications and opportunities for innovation, see Sect. 4.

To answer RQ3, we conducted an exploratory image-based perception study to examine perceived usefulness, anticipated real-world use, and expected comfort of representative XR-textile interaction concepts. Participants viewed concept illustrations derived from the design space and rated them using Likert-type scales, supplemented by open-ended feedback. The results indicate that familiar on-body forms were perceived relatively favorably within the presented concept set, while some less familiar and biodata-related concepts raised anticipated comfort and privacy concerns, see Sect. 5.

The Sect. 2 scoping review establishes the empirical foundation by identifying how smart textiles have been used in XR research to date. Building on this evidence base, the Sect. 4 design space abstracts recurring patterns and underexplored combinations into a framework for design reasoning. The two case studies then demonstrate how this framework can be applied in concrete XR design contexts. Finally, the Sect. 5 user study provides an initial user-centered exploration by examining how representative textile forms and interaction modes from the design space are perceived in terms of usefulness, anticipated real-world use, and expected comfort at the concept level.

This work advances XR and smart textiles in HCI through three contributions: (1) a scoping review synthesizing trends, gaps, and integration patterns; (2) a novel design space framed by Textile Form, Interaction Mode, and Application Domain, empowering researchers and designers to uncover synergies and innovate; and (3) exploratory empirical insights into how users perceive representative XR-textile interaction concepts, highlighting familiar interaction expectations and potential mismatches between perceived affordances and technological characteristics.

2 Scoping review of smart textiles in XR research

2.1 Scope and definitions

The aim of this review is to examine how smart textiles can be integrated with XR technologies. For this scoping review, we have considered XR and smart textiles as the two primary criteria.

XR is an umbrella term that encompasses various technologies such as VR, AR, and MR (Stanney et al. 2021; Vasarainen et al. 2021). VR technology enables users to fully immerse themselves in a simulated virtual world and interact with it. However, AR and MR allow users to experience both virtual and real-world elements simultaneously. However, AR generally overlays digital content onto the physical environment, whereas MR places greater emphasis on spatially registered interaction between physical and virtual elements. Our focus in this review is to explore the applications and implications of XR interaction when combined with smart textiles.

Smart textiles are textiles that have been engineered to include electronic components. These textiles can sense, process, or respond to environmental stimuli or user needs. They can be used in various settings, including traditional clothing, wearable accessories, home furnishings, and industrial applications, as long as the material is textile-based (Ghahremani Honarvar and Latifi 2017). We excluded non-textile wearables, such as smartwatches and headphones, because their primary interface structures are rigid rather than textile-based.

2.2 Methodology

Protocol, databases, & search Following the PRISMA-ScR (Tricco et al. 2018), we developed a systematic search protocol to identify relevant studies at the intersection of textile and XR technologies, as can be seen in Fig. 1. The review protocol was developed by the authors prior to screening and data extraction; however, it was not formally registered in a public repository. We acknowledge this as a limitation in methodological transparency.

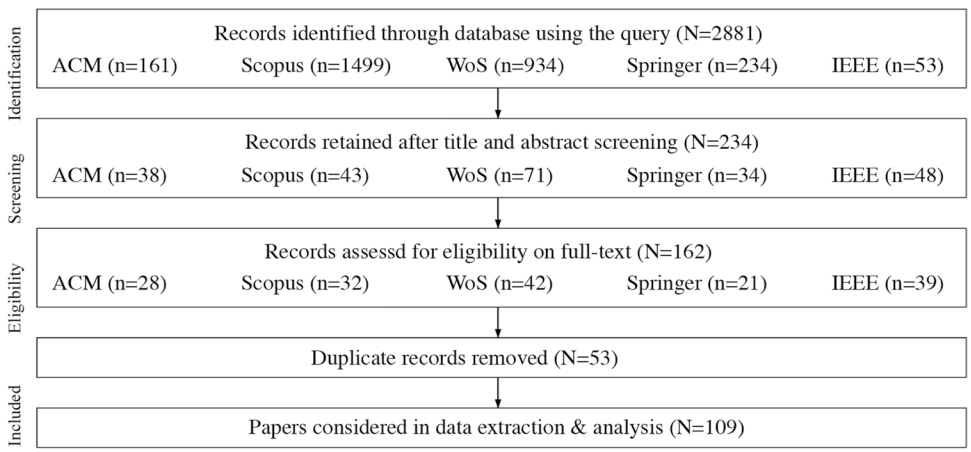


Fig. 1 The scoping review process and results in a PRISMA flow diagram

The search was conducted across the academic databases: Association for Computing Machinery (ACM) Digital Library, Scopus, Web of Science (WoS), IEEE (Institute of Electrical and Electronics Engineers) Xplore and Springer. The search was performed in September 2025.

The search query was constructed to encompass two keyword sets in the article title (TI): (1) textile-related keywords and (2) XR-related keywords. Taking WOS, for example:

TI=((textile* OR e-textile* OR wear* OR cloth* OR garment* OR suit* OR shirt* OR glove* OR sleeve* OR thread* OR knit* OR scarf* OR hat* OR string* OR pant* OR shoe* OR pillow* OR fashion* OR legging* OR curtain* OR quilt*) AND (VR OR AR OR MR OR XR OR "augmented reality" OR "virtual reality" OR "mixed reality" OR "extended reality")). IEEE Xplore was additionally searched using the same two conceptual keyword sets as the primary search. Due to IEEE Xplore's limit on the number of wildcard terms permitted in a single command search, the textile-related keyword set was divided into two otherwise equivalent search expressions, while the XR-related keyword set remained unchanged. The results of the two searches were merged and deduplicated before screening.

We restricted the search to article titles to improve precision at the intersection of smart textiles and XR, as broader title/abstract/keyword searches produced a large number of loosely related records in preliminary trials. The search yielded 2881 records (ACM = 161, Scopus = 1499, WoS = 934, Springer = 234, IEEE = 53).

Screening The screening process involved two stages. In the first stage, titles and abstracts were reviewed against the inclusion criteria (as discussed in Sect. 2). Exclusion criteria included non-textile wearables, unrelated domains, or non-empirical works. The second stage involved a full-text review of the remaining records to determine their eligibility for inclusion in the scoping review. Duplicates were removed during this process, resulting in a final corpus of 109 publications based on the search conducted in September 2025.

Data Extraction Eligible records underwent a structured data extraction process. A data extraction form was designed to capture key information, including publication year, venue, XR type (VR, AR, MR), research type (e.g., quantitative, qualitative, design & demonstration), smart textile form (e.g., gloves, suits), interaction modes, and application domains. The form was pilot-tested on a subset of papers and refined to ensure the consistency and completeness of data extraction. Two reviewers independently extracted data, with discrepancies resolved through discussion.

Synthesis Data synthesis involved organizing the extracted data into thematic categories informed by the research objectives and the content of the included studies. Categories included corpus characteristics, textile forms, interaction modes, and application domains. The synthesis aimed to provide a comprehensive overview of the current state of research at the intersection of smart textiles and XR technologies, identifying trends, gaps, and opportunities for future work.

2.3 Corpus characteristics

Publication time & venues After screening titles and abstracts, assessing eligibility, and removing duplicates, a final corpus of 109 papers was established, as shown in Fig. 1. The distribution of publication years in the corpus illustrates the surge in development of the field, with 18 papers from 2002 to 2017 and 91 from 2018 to 2025 (see Fig. 2a). The papers come from 49 different venues. We show the venues with at least two included papers in Fig. 2b. Among the 109 papers, 72 are conference papers, and 37 are journal papers (see Fig. 2c).

Contribution area Regarding the contribution areas of the papers, most of this research is concentrated in HCI, accounting for 84 papers. Other areas include materials (n=13), engineering (n=13), and clinical research (n=1). Figure 2d shows this distribution. Note that some papers contribute to multiple areas, leading to a sum exceeding the total corpus size.

XR Type Upon categorizing the literature based on the type of XR technology discussed, we tagged 70 papers solely focused on VR, 18 on AR, and four exploring both VR and AR. Additionally, 13 papers were dedicated to the MR domain, and four broadened their investigation to encompass the wider XR field (Hamari 2021;

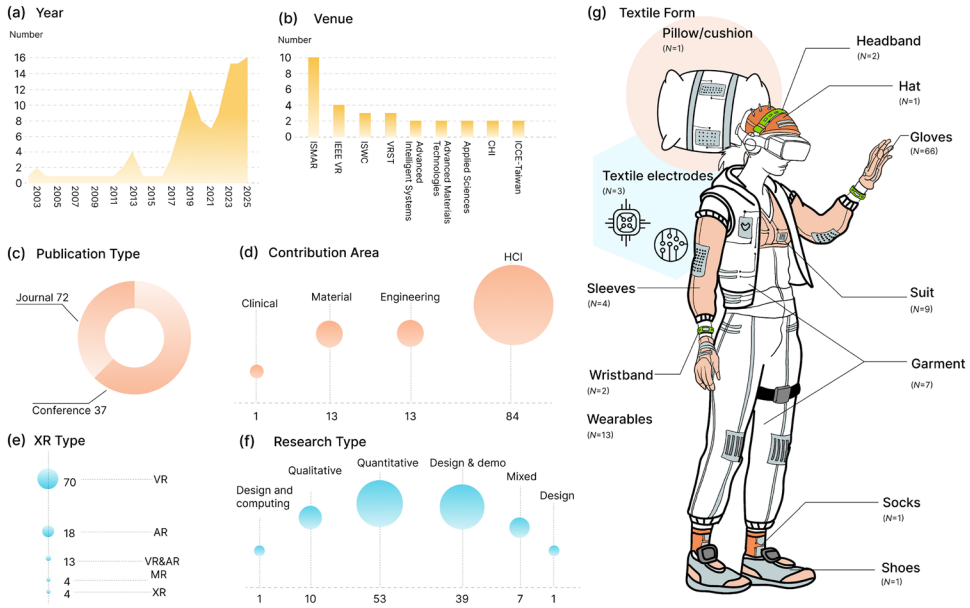


Fig. 2 Results of the scoping review: **a** Publication year, **b** publication venue (the top 8), **c** publication type, **d** contribution area, **e** XR type, **f** research type, and **g** smart textile form

Villarreal-Narvaez et al. 2025; Najm et al. 2024). Figure 2e summarizes the distribution of papers based on XR type.

Research type We also categorized the research types into quantitative analysis (n = 53), qualitative analysis (n = 10), mixed-methods analysis (n = 7), design & demonstration (n = 39), design (n = 1), and design & computing (n = 1), see Fig. 2f. Some studies employed more than one research approach, research-type categories were not mutually exclusive.

Smart textile type We categorized 11 types of smart textiles from the included research: gloves (n = 63), wearable (n = 15), suits (n = 10), garments (n = 9), sleeves (n = 3), textile electrodes (n = 3), wristbands (n = 2), socks (n = 1), headband (n = 1), hat (n = 1), and pillow/cushion (n = 1). We visualize them in Fig. 2g.

2.4 Summary

To answer RQ1, this scoping review of 109 publications reveals that smart textiles primarily serve as on-body input and output devices, with gloves dominating (n = 63) for haptic feedback, motion tracking, and gesture recognition, especially in VR applications. They enable intuitive, embodied interactions by translating physical movements into digital commands and delivering tactile or thermal sensations, enhancing immersion in domains like medical rehabilitation and industrial training.

The corpus highlights an interdisciplinary field led by HCI (84 papers), with contributions from materials science, engineering, and clinical domains. Research approaches are diverse, including quantitative evaluations (n = 53), design demonstrations (n = 39), and qualitative studies, reflecting a multifaceted exploration of textile-XR integration. While VR dominates, AR/MR and near/off-body textiles remain underexplored, signaling opportunities for multimodal and ambient interactions.

These findings underscore strengths in haptic-enabled VR while revealing gaps in AR/MR and non-on-body forms, providing an empirical foundation for deriving a comprehensive design space in the next section.

3 Extending the design space of smart textiles for XR

To address RQ2 and enable efficient navigation of this design space, we delineate it into three core dimensions: **Textile Form**, **Interaction Mode**, and **Application Domain**. By synthesizing data extracted from the literature, we map the design space, highlighting key trends and untapped opportunities for innovation.

3.1 Textile form

The textile form is a fundamental aspect of smart textile design for XR applications, which dictates the way users interact with and experience immersive environments. Textile forms can be classified according to their proximity to the user's body according to Thad Starner's research (Starner 2002): **On-body**, **Near-body**, and **Off-body**, each serving distinct functions and offering unique design opportunities.

3.1.1 On-body textiles

On-body textiles are worn directly by the user and are integral to the XR interaction. **Gloves**, **sleeves**, and **wristbands** are three hand- or arm-based smart textiles integrated into XR interactions. **Gloves** equipped with smart textiles have been researched extensively for their ability to provide haptic feedback and enable precise hand tracking in VR and AR applications. Notable work in this area includes (Adams et al. 2019; Ku et al. 2003; Piekarski and Smith 2006; Connelly et al. 2010; Thomas et al. 2011; Bazelle et al. 2018; Borja et al. 2018; Simon et al. 2014; Nanjappan et al. 2019; Hinchet et al. 2018; Alexandre et al. 2019; Adochiei et al. 2019; Liu et al. 2019; Cai et al. 2019; Wen et al. 2020; Zhu et al. 2020; Meng et al. 2020; Kim et al. 2022; Jamieson et al. 2023; Lee et al. 2023; Liu et al. 2023; Civelek and Fuhrmann 2022; Hsu et al. 2023; Wang et al. 2023; Boutin et al. 2024; Quaresma et al. 2018; Chan and Torah 2022; Gießer et al. 2023; Li et al. 2023; Yang et al. 2023; Zou et al. 2024; Dai et al. 2024; Hu et al. 2024; Jung et al. 2024; Seo et al. 2024; Vanichvoranun et al. 2024; Kim et al. 2025; Haikal et al. 2025; Lin and Cheng 2025; Cui et al. 2024; Sarkar et al. 2025; Thilakarathna and Phlernjai 2025; Hoang and Thomas 2012; Silva et al. 2013; Lee et al. 2017; Rumiński and Klinker 2018; Shor et al. 2018; De Tinguy et al. 2018; Masmoudi et al. 2021; Xia et al. 2022; Behnke et al. 2023; Yessentayev and Mukanova 2024; Huang et al. 2024; Pontreau et al. 2025; Kirby and Kim 2025; Guneser et al. 2025; Sasing et al. 2025; Nakao and Morimoto 2025; Lin and Liu 2025), showing the evolution of gloves from simple input devices to complex interaction tools that improve the XR experience.

Given XR's visual emphasis and tactile deficits, smart textiles often prioritize haptics, targeting sensitive areas like hands and arms where neural perception is heightened. This explains the prevalence of glove-based interactions.

Sleeves with integrated smart textiles offer a more focused approach to arm and forearm tracking, enabling gesture recognition and haptic feedback. Studies by (Narejo et al. 2020; Sousa et al. 2016; Chaudhari et al. 2024; Jung et al. 2024) have pushed the boundaries of what is possible regarding arm-based interactions in XR environments. **Wristbands** are popular for tracking wrist movements and providing localized haptic feedback in XR. They are also used for monitoring physiological signals (Houzangbe et al. 2018; Qian et al. 2020).

Clothes are another sector of on-body textiles. Smart **garments**, such as jackets or vests, have been explored for their potential to integrate multiple sensors and actuators, facilitating full-body interaction and feedback. As seen in (Häkkilä et al. 2017a, b; Ta et al. 2018; Priebe et al. 2020; Chong et al. 2021; Kiuchi et al. 2023; Jung et al. 2024; Cui and Mousas 2021; Muthukumarana et al. 2022), research contributions in this category highlight the transition from traditional clothing to interactive wearables that adapt to user movements and preferences. **Suits** can integrate multiple sensors and actuators for complex full-body interactions (Hamari 2021; Kang et al. 2023; Fitzgerald et al. 2007; Damian et al. 2013; Mangili and Netti 2024; Saladtook et al. 2025; Oota et al. 2019; Kishishita et al. 2019; Chen et al. 2021; Kammerlander et al. 2021). These suits can track movements and provide feedback in XR experiences.

Accessories are another type of on-body textile. For example, **socks** can track foot movements and provide feedback for balance and gait analysis within XR environments (Zhang et al. 2020). Head-worn textiles, including **hats**, **headbands**, and **helmets**, are used for head tracking, environmental interaction, and protection. Notable research exploring head-worn textiles in XR includes (Sakata et al. 2021), which explores integrating advanced materials and electronics for improved user experiences. On the other hand, **electrodes** can be integrated into various on-body textiles for biofeedback, monitoring physiological signals (Strachan et al. 2023; Rao et al. 2023; Li et al. 2023).

In this work, we use the term **wearables** to encompass a broad range of on-body smart devices, the textile form of which is hard to define. These wearables are designed to track movement, provide feedback, or serve as input devices (Hamari 2021; Häkkilä et al. 2017b; Ardaiz et al. 2019; Kuo et al. 2023; Cannavò et al. 2018; Kačerová et al. 2022; Liang et al. 2023; Najm et al. 2024; Jung et al. 2024; Villarreal-Narvaez et al. 2025; Duarte and Rodriguez 2025; Cheok 2002; Chu et al. 2012; Li et al. 2023; Yu et al. 2024).

On-body textiles exemplify smart textiles' versatility in XR, embedding sensing and actuation into attire for novel interactions. Beyond garments, accessories like **underwear** and **scarves** promise intimate, personalized XR engagements. These accessories highlight the untapped potential of smart textiles to transform common wear into interactive platforms.

3.1.2 Near-body textiles

Although not directly worn, near-body textiles are positioned in the user's immediate environment and play a significant role as interactive elements within the XR space. Empirical gaps in AR/MR integration limit their use, often due to synchronization challenges with virtual overlays.

For example, **pillows** can provide comfort and facilitate interaction in seated or reclined VR scenarios, as demonstrated by the innovative work on PillowVR (Kwon et al. 2019). When equipped with sensors, **cushions** can serve as interactive surfaces that offer tactile feedback, enriching the user's physical interaction with the XR space. Bedding items such as **quilts** (Kuusk et al. 2013) and **blankets** can envelop users in a warm and immersive environment, particularly suited for relaxation or gaming experiences. **Carpets** integrated with tracking technology can monitor user movement and deliver tactile feedback, making ground-based interactions in XR more intuitive and responsive. When fitted with sensors and actuators, **chair covers** can track posture and provide feedback, potentially improving ergonomics and user comfort during extended XR sessions. Furthermore, **yoga mats**, commonly used for exercise and relaxation, can be enhanced to track body position and movement, offering a new dimension for fitness applications within XR. Though underexplored, these textiles hold potential to expand XR toward holistic, ambient experiences.

3.1.3 Off-body textiles

Off-body textiles integrate environments into XR, activating surroundings as immersive participants. For example, **curtains** can serve as visual/auditory barriers providing cues or delineating spaces. **Nets** introduce dynamic elements in sports simulations. **Artistic installations** become interactive exhibits. **Table coverings** facilitate collaborative engagements. This approach invites a more ambient and holistic interaction with immersive technologies, where the user's entire setting can become a canvas for XR applications.

3.2 Interaction mode

Smart textiles in XR enable a bidirectional interaction paradigm: User input, Sensory output, Interaction interface and Multi-modal interaction. This section deconstructs these layers through a synthesis of 109 studies.

3.2.1 User input interaction: capturing human data for XR

Smart textiles mediate between human physicality and XR environments, translating subtle bodily expressions into computational commands. This bidirectional translation operates through two primary input modalities: **Motion-based input** and **Biodata-driven input**, each addressing distinct layers of user intentionality and physiological states. Motion-based interaction leverages textile-embedded sensors to detect user movements at varying scales:

- **Micro-gestures** Touch interaction utilizes smart textiles to discern and respond to physical contact. Extensive research (Ardaiz et al. 2019; Narejo et al. 2020; Yang et al. 2023; Ta et al. 2018; Wang et al. 2023; Zhu et al. 2020; Nanjappan et al. 2019; Adams et al. 2019; Jamieson et al. 2023; Hsu et al. 2023; Cai et al. 2019; Kang et al. 2023; Li et al. 2023; Wen et al. 2020; Thomas et al. 2011; Bazelle et al. 2018; Ku et al. 2003; Borja et al. 2018; Sakata et al. 2021; Quaresma et al. 2018; Connelly et al. 2010; Alexandre et al. 2019; Villarreal-Narvaez et al. 2025; Hoang and Thomas 2012; Rumiński and Klinker 2018; Shor et al. 2018; De Tinguy et al. 2018; Masmoudi et al. 2021; Kirby and Kim 2025; Sasing et al. 2025; Nakao and Morimoto 2025; Lin and Liu 2025) has been dedicated to integrating touch-sensitive textiles, emphasizing the development of interfaces that translate tactile inputs into digital interactions. Mid-air interaction allows users to engage with XR environments without the need for physical contact (Qian et al. 2020; Wang et al. 2023; Zhu et al. 2020; Nanjappan et al. 2019; Adams et al. 2019; Jamieson et al. 2023; Hsu et al. 2023; Gießer et al. 2023; Zou et al. 2024; Piekarski and Smith 2006; Hinchet et al. 2018; Meng et al. 2020; Lee et al. 2023; Way et al. 2019; Alexandre et al. 2019; Adochiei et al. 2019; Chan and Torah 2022; Boutin et al. 2024; Kirby and Kim 2025). For example, capacitive textiles in gloves, such as Yang et al.'s fully flexible smart gloves (Yang et al. 2023), track finger flexion using stretchable wires and sensors for VR object manipulation. Sleeves with bend sensors (Narejo et al. 2020) measure elbow range of motion (RoM) through inertial measurement units (IMUs), achieving 95% agreement with clinical goniometers.
- **Macro-gestures** Hand gesture interaction cases recognize and interpret movements and gestures, a focus of research by Wen et al. (2020); Thomas et al. (2011); Liu et al. (2019); Lin et al. (2022); Zou et al. (2024); Dai et al. (2024); Chaudhari et al. (2024); Hoang and Thomas (2012); Silva et al. (2013); Lee et al. (2017); Xia et al. (2022); Yessentayev and Mukanova (2024); Guneser et al. (2025); Lin and Liu (2025). Besides, foot/gait interaction studies track foot movements and gait (Zhang et al. 2020), and head interaction utilizes head movements for control and interaction, as discussed in Alexandre et al. (2019); Kwon et al. (2019); Li et al. (2023). RFID-enhanced gloves (Thomas et al. 2011) detect tangible button rearrangements in spatial AR environments, while pressure-sensitive carpets (Zhang et al. 2020) map footstep patterns for VR navigation.
- **Full-body interaction** Body motion interaction detects full-body movements (Cannavò et al. 2018), and social contact interaction enables interaction through physical contact, as seen in Ardaiz et al. (2019); Cheok (2002). Furthermore, pressure distribution is monitored across the textile surface (Wen et al. 2020). Besides, motion tracking is vital for creating a spatially aware experience where the XR system can respond to the user's physical presence and actions (Narejo et al. 2020; Yang et al. 2023; Ta et al. 2018; Chan and Torah 2022; Cannavò et al. 2018; Gießer et al. 2023; Fitzgerald et al. 2007; Kiuchi et al. 2023; Kuo et al. 2023; Kačerová et al. 2022; Damian et al. 2013; Wang et al. 2023; Hu et al. 2024; Jung et al. 2024; Seo et al. 2024; Jung et al. 2024; Kim et al. 2025; Haikal et al. 2025; Lin and Cheng 2025; Cui et al. 2024; Villarreal-Narvaez et al. 2025; Sarkar et al. 2025; Thilakarathna and Phlernjai 2025; Cheok 2002; Chu et al. 2012; Oota et al. 2019; Kishishita et al. 2019; Chen et al. 2021; Kammerlander et al. 2021; Behnke et al. 2023; Yu et al. 2024). Motion capture suits (Damian et al. 2013) integrate Xsens MVN systems to control AR avatars, and pneumatic gloves (Connelly et al. 2010) assist stroke rehabilitation by simulating grasp-and-release motions in VR.

Biodata-based interaction leverages physiological signals to create a responsive and personalized experience (Rao et al. 2023; Huang et al. 2024; Yu et al. 2024; Li et al. 2025). This method involves using smart textiles

capable of detecting and interpreting a range of physiological responses, thereby providing a direct link between the user's body and the XR system:

- **Muscle activity** Electromyography (EMG) captures electrical signals from muscles, offering insights into muscle activity and user intent (Strachan et al. 2023). EMG-embedded armbands (Quaresma et al. 2018) capture forearm signals to predict grasp intent in VR rehabilitation, while soft robotic gloves (Li et al. 2023) combine IMUs with tendon-driven force feedback for motor recovery.
- **Autonomic responses** Galvanic Skin Response (GSR) measurements track stress or emotional states by monitoring changes in the skin's electrical conductivity(Strachan et al. 2023). Heart Rate Variability (HRV) provides a window into the user's physiological well-being by analyzing the variation in time intervals between heartbeats (Houzangbe et al. 2018). GSR headbands (Houzangbe et al. 2018) adjust VR horror game difficulty based on skin conductance. Breath-sensing (Duarte and Rodriguez 2025) is also used as a holistic and conscious bodily process in VR interactions, measured through wearable interfaces to enhance user engagement and mindfulness by synchronizing breathing patterns with immersive audiovisual responses. Additionally, temperature monitoring is another facet of biofeedback, where temperature changes can offer environmental or physiological feedback (Hamari 2021).
- **Neurological signals** Electroencephalography (EEG) could also be included in this category, as it measures electrical activity in the brain and can be integrated into smart textiles for advanced biofeedback applications. For example, fabric electrodes integrated into VR headsets (Strachan et al. 2023) acquire EEG and GSR data for emotion-aware XR environments, though signal stability during prolonged use remains undocumented.
- **Eye/Gaze tracking** Multimodal systems such as Zou et al. (2024) combine inertial measurement units (IMUs) with gaze tracking to predict human handover intentions in AR-based industrial tasks, utilizing eye movement patterns to enhance interaction precision.

Key limitations reported in studies include motion capture suits requiring external processing units (Damian et al. 2013), EMG drift in prolonged sessions (Li et al. 2023), and capacitive sleeves lacking standardized RoM protocols (Narejo et al. 2020). These challenges highlight the need for material innovations and standardized evaluation frameworks.

3.2.2 Sensory output interaction: embodying digital realities through textile-mediated feedback

By actively shaping XR experiences through multisensory feedback, smart textiles bridge the gap between virtual abstractions and embodied perception. This output paradigm operates across tactile, thermal, visual, and auditory modalities, each engineered to evoke specific perceptual illusions that deepen immersion (Jiang et al. 2024).

Tactile feedback is a key element in XR interaction, providing tactile sensations to the user, allowing for a more tangible connection with the virtual world (Cai et al. 2019; Kang et al. 2023; Li et al. 2023; Adochiei et al. 2019; Chan and Torah 2022; Boutin et al. 2024; Hamari 2021; Simon et al. 2014; Civelek and Fuhrmann 2022; Kim et al. 2022; Liu et al. 2019; Dai et al. 2024; Hu et al. 2024; Najm et al. 2024; Seo et al. 2024; Vanichvoranun et al. 2024; Jung et al. 2024; Kim et al. 2025; Sarkar et al. 2025; Saladtook et al. 2025; Rumiński and Klinker 2018; Shor et al. 2018; De Tinguy et al. 2018; Oota et al. 2019; Kishishita et al. 2019; Cui and Mousas 2021; Masmoudi et al. 2021; Muthukumarana et al. 2022; Pontreau et al. 2025; Kirby and Kim 2025; Guneser et al. 2025; Sasing et al. 2025; Nakao and Morimoto 2025; Lin and Liu 2025). Pneumatic systems dominate haptic interfaces. The ThermAirGlove (Cai et al. 2019) uses airbags and temperature chambers to simulate material textures in VR. Electrostatic adhesion films in the DextrES glove (Hinchet et al. 2018) replicate object stickiness during virtual grasping, with no quantitative stickiness metrics provided. Full-body feedback is achieved via Kang et al.'s pneumatic-acoustic suit (Kang et al. 2023), which combines air bladders and vibration motors to simulate collision forces in VR training scenarios.

In addition to tactile feedback, force feedback systems respond to user input with resistive forces, creating a dynamic and responsive interaction (Priebe et al. 2020; Lin and Cheng 2025). Priebe et al. (2020) developed a

compression garment embedded in shape memory alloy (SMA) that applies spatially varying pressure to evoke emotional responses during consumption of VR media. Pneumatic interaction (Lin and Cheng 2025) is used to drive the glove to achieve natural finger flexion and extension, providing active assistance for grasping and extending exercises for stroke patients in MR.

Thermo feedback alters temperature sensations to align with the XR environment (Dai et al. 2024; Cai et al. 2019). The ThermAirGlove (Cai et al. 2019) demonstrates temperature shifts to match virtual material properties.

Visual feedback is achieved by incorporating visual elements directly into the textile, such as LED arrays, providing users with a seamless blend of physical and digital information (Cannavò et al. 2018; Qian et al. 2020; Sousa et al. 2016; Liu et al. 2023; Fitzgerald et al. 2007; Cui et al. 2024; Duarte and Rodriguez 2025; Li et al. 2023; Rao et al. 2023; Huang et al. 2024). Furthermore, audio feedback enhances the auditory experience, with textiles capable of generating or transmitting sound to complement the immersive experience (Sousa et al. 2016; Liu et al. 2023; Fitzgerald et al. 2007; Cui et al. 2024; Duarte and Rodriguez 2025).

Collectively, these sensory feedback modalities establish a comprehensive sensory output framework where smart textiles not only relay digital data but also evoke emotions and responses comparable to real-world experiences, thereby redefining the boundaries between the digital and physical worlds.

3.2.3 Textile as XR interface: beyond conventional hardware

The interaction interface determines how users engage with digital content. The interaction of AR markers, which use visual markers for tracking and interaction, has been explored by researchers (Häkkilä et al. 2017a, b). This approach allows for the integration of digital information with physical textiles, creating an interactive platform that can be manipulated within AR environments.

Additionally, projection technology has been utilized to display information directly on the surface of textiles, as seen in the work by (Wen et al. 2020; Chong et al. 2021). This capability transforms the textile into an interactive display, offering a new visual feedback and data presentation medium. Voice command interaction enables hands-free control via natural language processing (Wang et al. 2023), improving accessibility in constrained scenarios. These modalities highlight smart textiles' role in redefining XR connections, fostering nuanced, efficient, and personalized ecosystems.

3.3 Application domain

The integration of smart textiles with XR technologies manifests across a spectrum of application domains. Through systematic analysis of 109 studies, we categorized applications by their maturity level, see Fig. 3.
Medical validated applications Medical rehabilitation dominates validated implementations (21.2% of studies), where textile-XR systems demonstrate measurable clinical outcomes in virtual surgery (Boutin et al. 2024), medical training, psychotherapy, and rehabilitation (Jamieson et al. 2023; Alexandre et al. 2019; Narejo et al. 2020; Fitzgerald et al. 2007; Sousa et al. 2016; Liu et al. 2023; Adams et al. 2019; Li et al. 2023; Borja et al. 2018; Quaresma et al. 2018; Connelly et al. 2010; Chan and Torah 2022; Gießer et al. 2023; Kuo et al. 2023; Lee et al. 2023; Adochiei et al. 2019; Hu et al. 2024; Jung et al. 2024; Chaudhari et al. 2024; Silva et al. 2013; Oota et al. 2019; Masmoudi et al. 2021; Yu et al. 2024; Li et al. 2025). They can monitor patient vitals, provide haptic feedback for physical therapy, and enhance the realism of medical simulations. For instance, Connelly et al. (2010) developed a pneumatic glove *PneuGlove* integrated with VR environments for post-stroke hand rehabilitation, demonstrating measurable improvements in finger extension and grasp-release capabilities compared to traditional therapies. Similarly, Li et al. (2023) designed a soft robotic glove with IMUs and motor-tendon systems for VR-based hand rehabilitation, providing haptic feedback that enhances patient engagement. These systems prioritize biocompatibility and sensor accuracy, with methodologies validated through controlled trials involving stroke survivors and spinal cord injury patients.

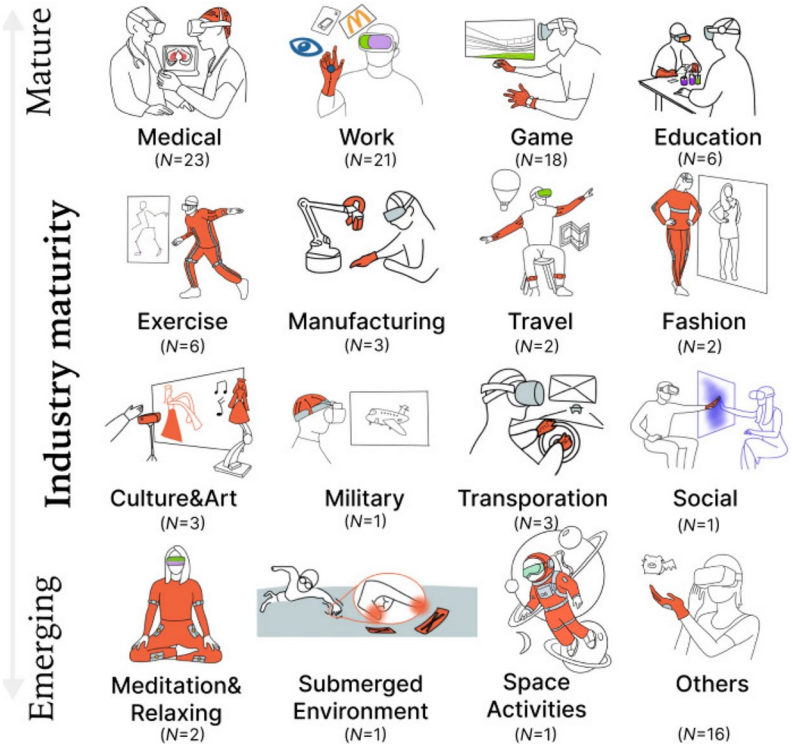


Fig. 3 The application domains of smart textiles for XR

Industrially deployed applications Smart textiles are being integrated into the workplace to assist with design (Kiuchi et al. 2023; Ta et al. 2018; Thomas et al. 2011; Hoang and Thomas 2012; Huang et al. 2024), programming (Kim et al. 2025), manufacturing (Zou et al. 2024), assistance (Kačerová et al. 2022; Huang et al. 2024), and remote collaboration (Chong et al. 2021; Sarkar et al. 2025; Lin and Liu 2025; Behnke et al. 2023) to enhance training efficiency and safety. Kačerová et al. (2022) utilized motion capture clothing and AR overlay to optimize ergonomic design in the workplace, iteratively optimizing production line layout through virtual simulation, and reducing the need for physical prototypes. Nanjappan et al. (2019) explored fabric-based wearable devices as an interactive interface for handheld AR driving, and determined the interactive vocabulary of wrist and thumb index finger touch gestures through user-guided research. In outdoor scenarios, the conductive fabric gloves developed by Piekarski and Smith (2006) support 6-degree-of-freedom tracking in outdoor AR environments, and their low power consumption and wireless characteristics have been validated in actual maintenance tasks. Although some studies, such as Chan and Torah (2022) electronic textile gloves, have not quantified specific error-rate reductions, their design goals clearly target practical needs in industrial scenarios.
Prototype-validated emerging applications The application in education is still in the prototype verification stage, but has demonstrated unique potential (Wang et al. 2023; Meng et al. 2020; Ku et al. 2003; Civelek and Fuhrmann 2022; Hu et al. 2024; Haikal et al. 2025; Lin and Cheng 2025; Cui et al. 2024; Shor et al. 2018; Guneser et al. 2025; Sasing et al. 2025; Rao et al. 2023; Nakao and Morimoto 2025). Wang et al. (2023) integrated IMU and pressure sensors into their chemical experiment smart gloves, which guided students to complete sulfuric acid dilution experiments in VR. Meng et al. (2020) proposed a multimodal intention perception algorithm that combines gesture, speech, and visual data to improve experimental efficiency. The innovative case of social XR includes Kwon et al. (2019) PillowVR system, which uses pressure sensing pillows to achieve bed rest VR narrative interaction and focuses on children's bedtime scenes.

The gaming industry has embraced smart textiles and XR to enhance the player's immersion and interaction with the game environment. They can provide haptic feedback, track movements, and adapt to the game's narrative (Jamieson et al. 2023; Alexandre et al. 2019; Sakata et al. 2021; Kang et al. 2023; Hamari 2021; Ardaiz et al. 2019; Houzangbe et al. 2018; Hu et al. 2024; Duarte and Rodriguez 2025; Saladtook et al. 2025; Cheok 2002;

Chu et al. 2012; Lee et al. 2017; Kishishita et al. 2019; Cui and Mousas 2021; Chen et al. 2021; Xia et al. 2022; Yessentayev and Mukanova 2024). For example, Ardaiz et al. (2019) explored the physical contact and interaction mechanism between VR and real-world players in their wearable touch interface. Although specific indicators of social presence enhancement have not been quantified, preliminary user evaluations show its contribution to enhancing game physicality.

Frontier exploration in cross disciplinary challenges Underwater and cognitive enhancement applications are limited by material technology and are in the early experimental stage. The hydrophobic frictional electric fiber developed by Liang et al. (2023) achieved an underwater motion tracking accuracy of 89%, providing a technical foundation for marine safety training. Hsu et al. (2023) integrated fabric electrodes into VR headsets to monitor EEG and skin electrical responses, exploring the emotional perception XR environment, but its signal stability still needs to be verified in long-term use. Although the gaming industries have commercial potential, such as Zhu et al.'s tactile feedback gloves (Zhu et al. 2020), issues with washability and durability hinder their large-scale application. Cross-disciplinary technology transfer has begun to emerge, for example, Zhang et al. (2020) friction electric smart socks were originally designed for VR fitness games and have now expanded to user recognition scenarios in smart homes.

Niche application directions While mainstream applications of smart textiles in XR focus on healthcare, industry, and education, several specialized domains demonstrate innovative potential. In military training, smart textiles enable immersive flight simulations (Sakata et al. 2021) and scenario-based exercises with haptic feedback for personnel preparation. For sports and exercise, motion-capture suits (Kang et al. 2023; Cannavò et al. 2018) and outdoor activity wearables (Piekarski and Smith 2006) provide XR-enhanced coaching and performance analytics. The travel sector explores interactive navigation fabrics for tourists, while fashion pioneers AR-enabled garments (Häkkilä et al. 2017a, b) for virtual fitting and dynamic aesthetic displays. In cultural preservation, smart textiles drive mixed-reality installations (Way et al. 2019; Lin et al. 2022; Kammerlander et al. 2021) to digitize heritage crafts like glove puppetry. Social XR experiments with haptic quilts (Hamari 2021) for remote tactile communication (Yessentayev and Mukanova 2024), and meditation systems like PillowVR (Kwon et al. 2019) use pressure-sensitive textiles to create ambient relaxation environments. Emerging frontiers include material simulation gloves for VR texture rendering (Cai et al. 2019; Bazelle et al. 2018; De Tinguy et al. 2018) and adaptive garments with shape-memory alloys (Priebe et al. 2020), pushing the boundaries of embodied interaction in XR. VR is also used to simulate space activities (Mangili and Netti 2024), particularly Extra-Vehicular Activities (EVAs), by creating a cost-effective, flexible, and immersive environment that enhances realism through the use of a space suit mock-up.

The diverse application space of smart textiles in XR demonstrates their potential to transform various aspects of our lives, from enhancing daily activities to providing novel interactions and experiences.

3.4 Shaping the design space

The design space mapping involved categorizing the included studies based on textile form and interaction mode using the Zwicky Morphological Box (Zwicky 1967). This process enables visualization of research distribution, identification of activity clusters and isolated areas, and recognition of potential synergies across dimensions, as illustrated in Fig. 4. Application Domain was not shown in Fig. 4 as a core dimension because it functions as an orthogonal layer, allowing domains to span multiple form-mode combinations without constraining the primary structural synergies.

Shaping this design space employs a structured approach to explore and visualize synergies in smart textiles for XR. We first define core parameters: Textile Form (On-body, Near-body, Off-body) and Interaction Mode (User Input, Sensory Output, Interaction Interface), subdivided into categories from the review. The matrix in Fig. 4 represents these parameters as axes, with intersections highlighting feasible combinations and opportunities for innovation; slashes denote infeasible pairings. Mapping existing studies provides a snapshot of the field,

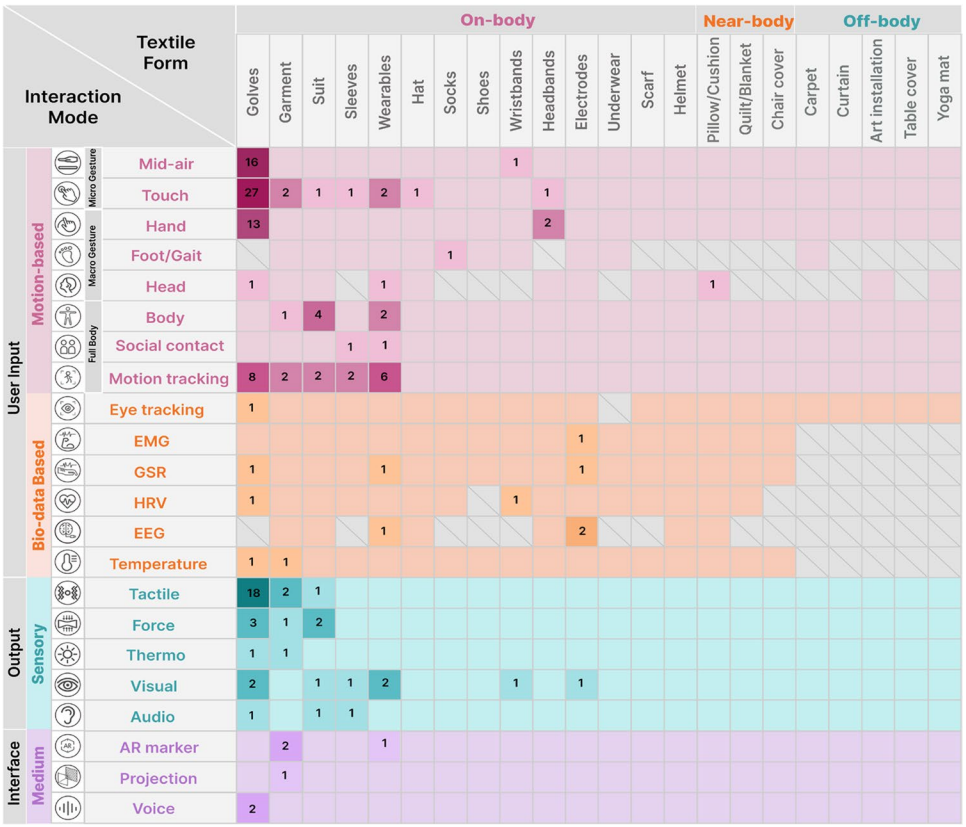


Fig. 4 The design space of smart textiles in XR

revealing gaps—such as near/off-body forms with multi-modal biodata—that guide future exploration and development in textile-XR integration.

4 Using the design space framework with design cases

4.1 Example of applying the design space framework

To effectively utilize this design space, it is essential to understand its structure, parameters, and innovation potential. The following steps guide its application:

(1) Understand the Design Space and Parameters Familiarize with the core dimensions—Textile Form and Interaction Mode—which reflect current research and capabilities in textile-XR integration.

(2) Identify the Research Gaps Map existing studies onto the design space to pinpoint well-explored areas (e.g., on-body haptic gloves) and underexplored ones (e.g., near-body multi-modal interfaces), directing focus toward innovation opportunities.

(3) Identify Application Scenario For a specific XR scenario, such as enhancing immersion in VR gaming or providing feedback in AR training, choose relevant parameters (e.g., near-body form with sensory output) and experiment with combinations to prototype solutions.

(4) Select Parameters & Explore Combinations Test prototypes iteratively, evaluating effectiveness and refining based on user feedback. The design space serves as a reference to maintain focus and explore promising avenues.

For example, consider the work of Häkkilä et al. (2017a), which explores the integration of AR markers into clothing for interactive applications. The design space could be applied here to identify the application scenario

(interactive fashion or gaming), textile form (clothing), and the interaction mode (AR marker interaction). The haptic stimulation glove (Boutin et al. 2024) aimed at providing haptic feedback and motion tracking for fine motor rehabilitation in VR environments. Using the design space, the application scenario is "rehabilitation", specifically for motor function recovery. The textile form is identified as "gloves", and the primary interaction mode is "haptic feedback" combined with "motion tracking".

By applying the design space, the team can systematically explore the possibilities of smart textiles in XR, leading to the development of novel and impactful products and applications. This design space not only aids in creating new technologies but also fosters a deeper understanding of the potential and limitations of smart textiles in XR environments.

4.2 Applications of the design space framework

Applying the design space is instrumental in creating innovative smart textile solutions for specific XR applications. This section explores the application of the design space through two cases.

4.2.1 Case 1: smart pillow for social interaction in VR

The smart pillow enables remote social interactions in VR, serving as a comfort object and interaction device for non-verbal communication, enhancing presence and intimacy when physical co-location is infeasible. Its design aligns with the framework's parameters.

Textile Form: As a near-body textile, the pillow prioritizes comfort and accessibility in seated or reclined positions.

Interaction Mode: It employs motion-based input via gesture sensing (e.g., hugging or squeezing) to mimic real-world cues, coupled with sensory output through haptic actuators for simulated touch. Additionally, integrated GSR sensors detect emotional states, adding biodata-driven responsiveness.

In operation, a user's hug transmits sensor data to the VR system, rendering corresponding virtual gestures and reciprocal haptic feedback to evoke mutual embrace. As shown in Fig. 5, this facilitates shared VR spaces where avatars mirror actions, fostering emotional connection in long-distance scenarios. This concept illustrates how a near-body textile interface could support non-verbal remote social interaction in VR.

4.2.2 Case 2: wearable textile keyboard sleeves for typing in MR

The wearable textile keyboard sleeves illustrate a speculative design possibility for text input in virtual/augmented spaces where traditional keyboards may be unavailable or undesirable. Their design draws from the framework's parameters.

Textile Form: As on-body sleeves worn on the forearms, they integrate conductive threads into flexible, conformable fabric for seamless wear.

Interaction Mode: They support user input via touch and gesture, with capacitive sensing registering keypresses as an intuitive tactile interface, eliminating physical keyboards.

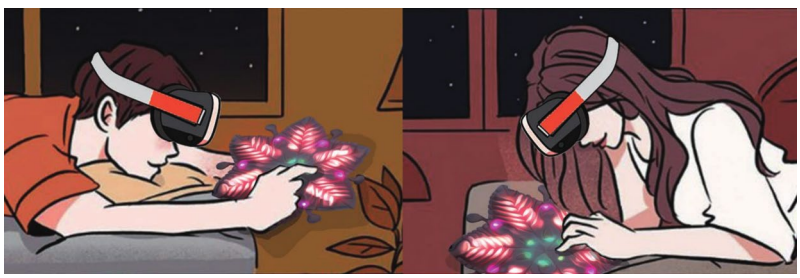


Fig. 5 The smart pillow for remote social interaction in VR

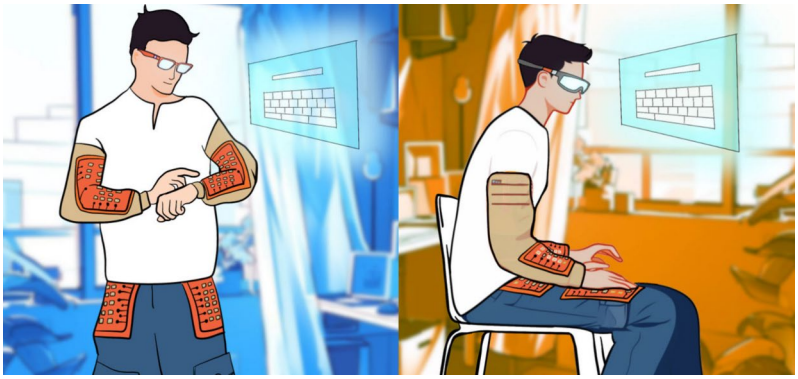


Fig. 6 The wearable textile keyboard sleeves for typing in MR



Fig. 7 Concept images used in the user study. Textile forms: **a** gloves, **b** garment, **c** suit, **d** sleeves, **e** wearables, **f** hat, **g** socks, **h** shoes, **i** wristbands, **j** headbands, **k** electrodes, **l** underwear, **m** scarf, **n** helmet, **o** pillow, **p** quilt, **q** chair cover, **r** carpet, **s** curtain, **t** art installation, **u** table clothes, and **v** yoga mat

In use, conductive patterns mimic a standard layout; finger contacts generate electrical signals for accurate registration, calibratable to user styles. In this concept, conductive textile regions could provide touch-based input while viewing MR displays, integrating into scenarios like virtual meetings, office tasks, or gaming.

Potential design benefits to be investigated include portability, a relatively large textile interaction area, and the possibility of leaving the fingers available for other manipulations. However, forearm-worn interfaces may also introduce trade-offs related to fit, reachability, pressure, thermal comfort, movement restriction, and prolonged wear. These factors require validation through functional prototypes and embodied user studies before any ergonomic advantage can be established (Fig. 6).

5 User perceptions of XR-textile interaction concepts

5.1 Study procedure and measurements

To answer RQ3 and examine how users perceive representative XR-textile interactions derived from the design space, we conducted an exploratory, image-based online perception study with 30 participants. The study employed a within-subject design, with the two primary dimensions of the design space—**Textile Form** and **Interaction Mode**—serving as the basis for the stimulus set. Importantly, the study was designed as an exploratory concept-perception study rather than an evaluation of functional textile-XR systems. Participants were shown simplified illustrations and brief annotations describing the intended interactions, but they were not provided with detailed technical instruction regarding sensing mechanisms, actuation principles, signal acquisition requirements, implementation constraints, or actual embodied sensations. Accordingly, the ratings represent participants' perceptions of the depicted concepts and their anticipated affordances. They should not be interpreted as informed assessments of technological feasibility, objective comfort, or the relative technical suitability of different technologies.

We generated a total of 44 concept illustrations: 22 representing textile forms and 22 representing interaction modes, each depicting users engaging in XR-related scenarios (see Figs. 7 and 8). The images were created using

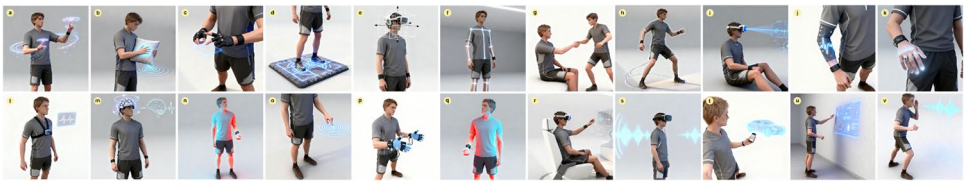


Fig. 8 Concept images used in the user study. Interaction modes: **a** mid-air, **b** touch, **c** hand, **d** foot, **e** head, **f** body, **g** social contact, **h** motion tracking, **i** eye tracking, **j** EMG, **k** GSR, **l** HRV, **m** EEG, **n** temperature, **o** tactile, **p** force, **q** thermo, **r** visual, **s** audio, **t** AR marker, **u** projection, and **v** voice

digital tools in a simplified schematic style, similar to prior concept-based design research (Jansen et al. 2022), and were annotated to clarify the intended interaction. The stimulus set was derived from the design space and included both forms commonly represented in the reviewed literature and some speculative extensions intended to probe underexplored opportunities. The images were randomized and presented one at a time to reduce order effects.

Thirty participants were recruited through the WeChat social platform in October 2025. Eligible participants were adults aged 18 or above and had prior XR experience, defined here as having used VR, AR, or MR systems at least once before the study. This study was designed as an exploratory perception study rather than a population-level validation study. The sample included 19 female and 11 male participants, aged 20–33 years. Using an asymptotic noncentral chi-square approximation for the Friedman test, a sensitivity power analysis was conducted for the within-subject design ($N = 30$, $k = 22$, $\alpha = .05$, power = .80). The analysis indicated that the study was sensitive to an omnibus effect corresponding to Kendall's $W \approx .067$ or larger. Accordingly, the study was intended to identify broad differences across the concept set rather than small pairwise differences between individual conditions.

The study began with informed consent and a short demographic questionnaire. Participants then viewed the concept images one by one in randomized order and rated each image on three 8-point Likert-type scales ranging from 0 to 7 (0 = strongly disagree, 7 = strongly agree): (1) perceived usefulness ("This interaction would be useful in real XR applications"), (2) anticipated real-world use ("I would use this in everyday XR scenarios"), and (3) expected comfort ("This interaction would feel comfortable to use"). In addition to the rating items, participants were invited to provide open-ended comments about each concept or about any textile forms and interaction modes they found particularly appealing, confusing, uncomfortable, or unrealistic.

The study was conducted online. On average, the full procedure took approximately 20 min per participant. The questionnaire and stimulus materials used in the study are provided in the Appendix.

5.2 Results

5.2.1 Textile form

The Friedman test revealed significant differences in ratings across the 22 textile forms for perceived usefulness ($\chi^2(21) = 71.28$, $p < 0.001$, Kendall's $W = .113$), anticipated real-world usage ($\chi^2(21) = 76.09$, $p < 0.001$, $W = .121$), and expected comfort ($\chi^2(21) = 98.85$, $p < 0.001$, $W = .157$). Kendall's W was reported as an effect-size measure for the Friedman tests, with larger values indicating greater agreement in the rank differentiation among conditions. Given the exploratory nature of the study, the effect sizes are reported primarily to characterize the magnitude of the observed omnibus differences rather than to support fine-grained comparisons between individual concepts.

Figure 9 shows the average ratings for perceived usefulness, with forms like gloves, shoes, wearables, and suits receiving higher scores within the illustrated concept set. These patterns describe participants' perceptions of the depicted concepts. In contrast, forms such as electrodes, curtains, tablecloths, and quilts scored lower, as

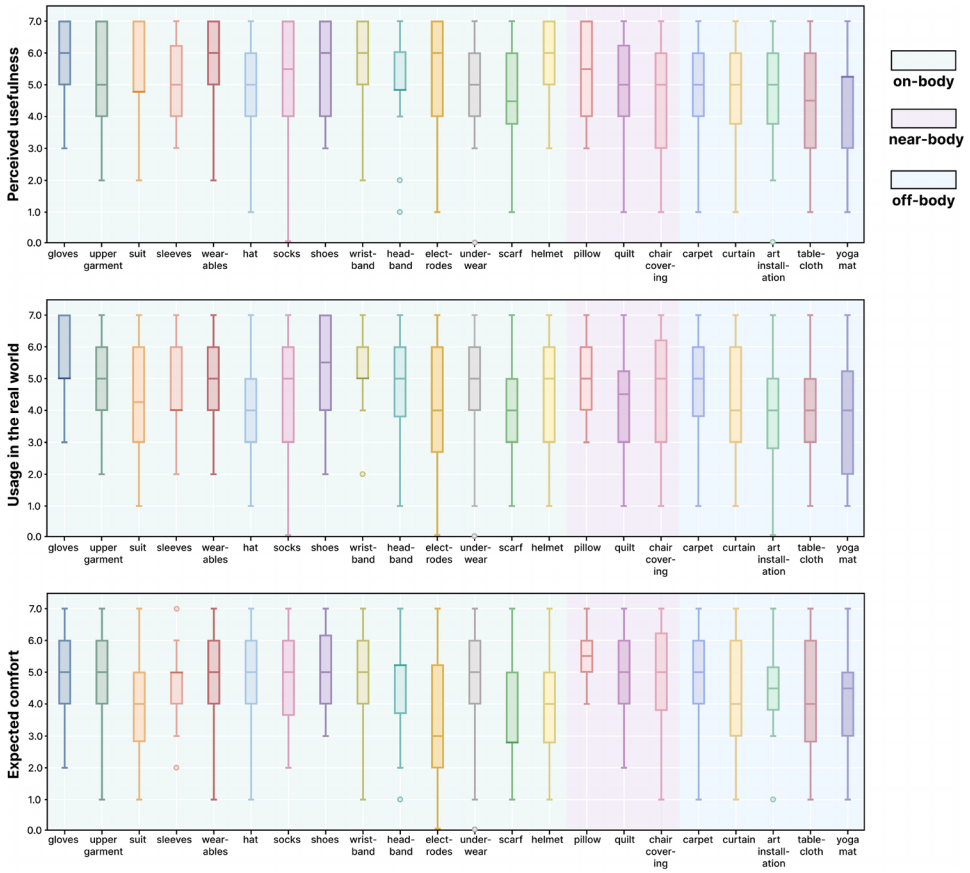


Fig. 9 Box plot of perceived usefulness, usage in the real world, and expected comfort ratings for textile forms in XR

participants struggled to envision their integration. Similar patterns emerged for real-world usage and comfort (Fig. 9), where on-body forms were generally favored over near- or off-body alternatives.

5.2.2 Interaction mode

Likewise, significant differences were observed across the 22 interaction modes for perceived usefulness ($\chi^2(21) = 75.44$, $p < 0.001$, Kendall's $W = .120$), anticipated real-world usage ($\chi^2(21) = 76.28$, $p < 0.001$, $W = .121$), and expected comfort ($\chi^2(21) = 101.84$, $p < 0.001$, $W = .162$). As illustrated in Fig. 10 for usefulness, modes like gesture interaction, visual, tactile, mid-air, and EEG received higher ratings within the illustrated concept set. Participants also expressed interest in thermal and force-feedback concepts. However, modes involving GSR and EMG scored lower on comfort, often due to perceived physiological discomfort, while usage patterns mirrored these trends.

5.2.3 Open feedback

Thematic analysis of open-ended responses highlighted skepticism toward certain textile forms, with participants (P3, P6, P12, P16, P19) expressing doubts about electrodes, curtains, tablecloths and quilts in XR contexts, leading to lower scores on usefulness and real-world use. Comfort concerns were noted for electrodes, scarves, wristbands, and headbands (P14, P17, P20). In contrast, most of the participants (P13, P15, P16, P18, P21, P22, P24, P26) praised gloves, shoes, wearables, and suits for their utility. For interaction modes, enthusiasm was evident for gesture, visual, tactile, mid-air and EEG (P6, P7, P9, P17, P19, P23, P25, P27), with anticipation for developments in EEG, mid-air, motion capture, thermal, and force feedback. A minority (P8, P10, P11, P12) viewed GSR and EMG as uncomfortable physiologically.

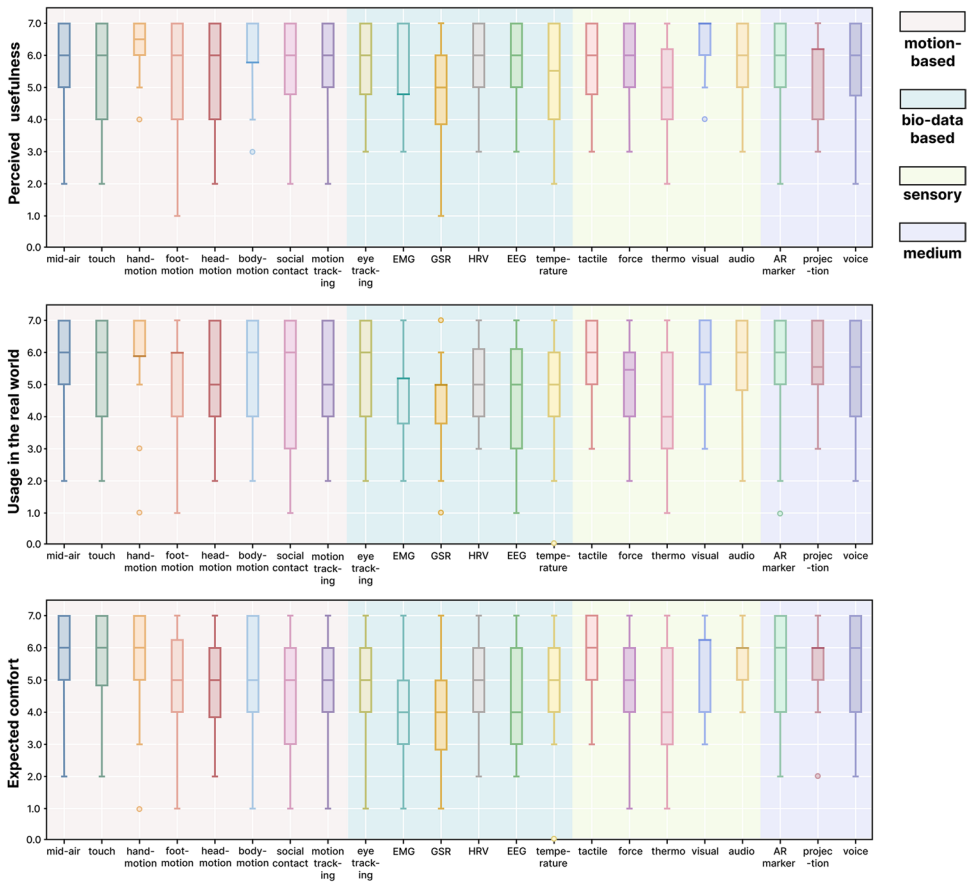


Fig. 10 Box plot of perceived usefulness, usage in the real world, and expected comfort ratings for interaction modes in XR

6 Discussion

6.1 Trends and gaps in XR-smart textile research

Our scoping review of 109 publications, combined with the user study findings, reveals key trends in the integration of smart XR textiles while highlighting persistent gaps. A dominant trend is the focus on VR applications, particularly using body forms like gloves for haptic feedback, as seen in medical rehabilitation and industrial training (Boutin et al. 2024; Zou et al. 2024). This aligns with broader HCI research on immersive interactions (Rauschnabel et al. 2022), where tactile modalities enhance embodiment (Jiang et al. 2024). However, gaps emerge in AR/MR explorations, and near-body & off-body textiles, e.g. pillows, blanket, remain sparsely represented in the corpus. The results of the user study provide a complementary user-perception perspective on this pattern: familiar body forms (gloves, suits) received high usefulness ratings (median>5), while off-body items like curtains received relatively low ratings, consistent with participants' difficulty envisioning their XR integration.

Bibliometric analyses from recent reviews indicate a surge in publications post-2018, driven by advances in material science and flexible electronics, yet commercialization lags due to gaps in unified standards and multi-user scalability (Liu et al. 2021). Emotion-related e-textiles show promise for social XR (Häkkilä et al. 2017a), but our qualitative feedback (e.g., P03's skepticism on quilts) highlights underexplored privacy and integration issues (Priebe et al. 2020).

6.2 Perceived affordances, mental models, and design implications

The exploratory study revealed systematic differences in how participants perceived the illustrated XR-textile concepts. These differences should be interpreted as concept-level perceptions shaped by familiarity, visual framing, and existing mental models. For textile forms, on-body options like gloves and suits garnered high ratings on usefulness and real-world usage (medians>5), reflecting their practicality in scenarios such as VR rehabilitation or MR typing, as evidenced by review trends where gloves dominate for haptic input (Connelly et al. 2010; Li et al. 2023). Participants' praise (e.g., P13-P26) for these forms aligns with their established roles in motion-based interactions (Yang et al. 2023; Narejo et al. 2020). Conversely, near- and off-body forms (e.g., pillows, curtains) scored lower, with skepticism from P03-P19 on envisioning applications like ambient feedback, underscoring review gaps in non-wearable textiles. These patterns suggest that familiarity may influence how readily users can imagine particular textile forms in XR, but they do not establish the inherent superiority of on-body forms.

For interaction modes, established methods like gesture and tactile received relatively favorable ratings on comfort and usage, consistent with their prevalence in gaming and education domains (Wang et al. 2023; Ardaiz et al. 2019). Enthusiasm for EEG and thermal (P06-P27) suggests potential for biofeedback innovations, as in our design space's multimodal examples (Strachan et al. 2023; Cai et al. 2019). However, lower comfort scores for GSR and EMG (P08-P12) due to physiological concerns mirror challenges in biodata inputs from the review (Houzangbe et al. 2018; Quaresma et al. 2018), implying the need for improved user experience in electrode devices through textile electrodes (Cai et al. 2019). Overall, higher acceptance for on-body and tactile elements reinforces user-centered refinements, bridging review trends with practical XR applications.

An interesting finding from the user study is that EEG was perceived more positively than EMG and GSR, despite the latter often being technically simpler or less obtrusive in practice. We interpret this as a perception effect shaped by the visual framing of the concepts and by users' current mental models of XR hardware. EEG may have appeared more acceptable because participants associated it with head-mounted XR systems and therefore imagined it as part of an existing device ecosystem. By contrast, EMG and GSR may have been interpreted as requiring tighter skin contact, explicit bodily instrumentation, or medically oriented monitoring, which could evoke discomfort, intrusiveness, or privacy concerns. Accordingly, the relatively positive perception of EEG should not be interpreted as evidence that EEG is technically simpler, more comfortable, or more suitable for XR. Rather, the discrepancy illustrates that perceived affordance and technological reality may diverge when emerging interfaces are evaluated through concept representations alone.

The design space offers actionable implications for advancing XR-textile interactions. For instance, the predominance of on-body forms like gloves in VR applications suggests prioritizing ergonomic designs to enhance user acceptance, where study participants rated gloves highly on usefulness and comfort due to perceived practicality. Extending this to near-body textiles, such as pillows for social VR as in our case study, implies opportunities for ambient, emotional interactions that address gaps in multi-modal feedback, like combining haptic output with biodata input to simulate remote presence. User feedback highlighting privacy concerns for physiological modes (e.g., EEG, GSR) further implies the need for transparent, user-configurable systems, aligning with review findings on underexplored biofeedback in XR.

6.3 Future research trends

6.3.1 VR trends: enhanced immersion through haptic and sensory feedback

Recent work demonstrates growing interest in combining tactile, force, and thermal feedback to enrich embodied VR experiences. For example, ThermAirGlove integrates thermal and force feedback to support virtual material perception (Cai et al. 2019). Health-focused VR wearables have also explored physiological sensing, including GSR/EEG, for adaptive or affect-aware experiences (Strachan et al. 2023; Houzangbe et al. 2018). Trends include lighter headsets with textile sensors for gaming/entertainment, reducing commercialization lags in washability

(Sakata et al. 2021). In-vehicle parallels suggest VR textiles for driver training, using actuators for vestibular stimuli in vehicles (Jansen et al. 2022), expanding to metaverse rehab where textiles simulate real-world sensations.

6.3.2 AR trends: fashion tech and virtual try-ons

AR in smart textiles is shifting to consumer fashion, extending review's visual modes, e.g., AR markers on clothing (Häkkilä et al. 2017a, b). Recent studies explore virtual try-on and AR-assisted garment design (Ta et al. 2018; Kiuchi et al. 2023). This addresses AR gaps (n=18 papers), with textiles like e-textiles enabling real-time pattern changes viewable in AR applications. AR-guided therapy sleeves illustrate opportunities to combine movement feedback with wearable sensing in rehabilitation contexts (Narejo et al. 2020; Sousa et al. 2016). The relatively favorable ratings of gesture and visual concepts for AR navigation suggest that these familiar interaction metaphors may be readily understood by participants. Besides, sustainability trends feature eco-textiles with AR interfaces, like projection on tablecloths for interactive maps, fostering ambient AR in education/gaming.

6.3.3 MR trends: hybrid real-digital interfaces

MR textiles trend toward metaverse/spatial computing, filling the corpus gaps (MR n=13) with hybrid forms like gloves for MR typing (Lin and Cheng 2025; Cui et al. 2024). Industrial MR with biofeedback, extending in-vehicle actuators (Jansen et al. 2022), includes EMG/GSR in wearables for ergonomic MR training (Quaresma et al. 2018; Li et al. 2023). Physiological sensing in MR/XR, including EEG/GSR-based approaches, also raises questions around privacy, signal reliability, and embodied comfort (Strachan et al. 2023). Ambient MR for daily life, like smart textiles in vehicles or homes, leverages projection/visual modes for collaborative spaces (Wen et al. 2020), with sustainable energy-harvesting for prolonged use (Zhang et al. 2020), aligning with education domains (Wang et al. 2023).

6.3.4 Underexplored textile directions beyond haptic gloves

While haptic gloves dominate the current literature, several alternative textile directions remain comparatively underexplored for the next phase of XR research. First, **sleeves and forearm-worn textiles** provide a relatively large surface for integrating sensing and actuation while leaving the fingers available for direct manipulation. Compared with gloves, they interfere less with fine hand dexterity while still supporting gesture sensing, muscle activity capture, and localized haptic feedback. However, forearm-worn systems may introduce trade-offs related to fit, pressure, thermal comfort, restriction, and prolonged wear. Sleeves should therefore be regarded as a context-dependent design opportunity rather than a universally preferred alternative to gloves. Second, **garments and vests** can enable more distributed, full-body interaction, making them suitable for posture guidance, affective feedback, training, and rehabilitation. Their larger textile area also supports multimodal integration of sensing and actuation. Third, **near-body textiles** such as pillows, cushions, blankets, or chair covers may be particularly valuable in social, therapeutic, and comfort-oriented XR contexts, where the goal is not precise manipulation but emotional communication, bodily grounding, or long-duration support. Finally, **ambient and environmental textiles** such as carpets, mats, and curtains, though underexplored, may open new possibilities for spatial interaction and peripheral feedback in AR/MR environments. These alternatives are noteworthy not because they replace gloves, but because they support different interaction goals.

Table 1 Interdisciplinary advances enabling smart textiles for XR

Layer	Example technologies	Potential XR contribution	Key challenges
Materials	stretchable conductors, textile electrodes, nanomaterials	richer sensing and haptics	durability, skin compatibility
Fabrication	embroidery, knitting, lamination	scalable textile integration	repeatability, washability
System integration	power, wireless modules, sensor-actuator coupling	stable real-time interaction	bulk, maintenance
AI/ML	multimodal fusion, adaptive feedback	personalization and prediction	privacy, explainability

6.3.5 Opportunities brought by interdisciplinary advances

Advancing smart textiles for XR requires tighter collaboration across HCI, textile engineering, materials science, electronics, computer science, and fashion design. Such interdisciplinary work is necessary because textile-XR systems must simultaneously address sensing, actuation, wearability, aesthetics, and user experience.

At the material level, emerging technologies such as stretchable conductors, textile electrodes, thermally responsive fibers, soft pneumatic structures, and nanomaterial-based textile components can expand tactile, thermal, and physiological interaction channels while improving wearability (Wang et al. 2023, 2021). At the fabrication level, embroidery, knitting, coating, lamination, and hybrid assembly methods make it increasingly feasible to integrate sensing and actuation into textile substrates without sacrificing softness, breathability, and flexibility, see Table 1.

At the system level, future progress depends on more robust integration of sensors, actuators, power, and communication modules into textile systems that can tolerate repeated deformation, maintenance, and long-term use. Addressing commercialization issues such as durability, washability, sensor integration, and sustainability may further accelerate adoption in domains such as education, gaming, rehabilitation, and remote collaboration (Wang et al. 2023; Ardaiz et al. 2019; Liu et al. 2021).

In parallel, AI and machine learning can extend textile-XR systems beyond fixed input–output mappings by enabling multimodal inference, adaptive feedback, and personalized interaction (Qiu et al. 2024). By combining motion, touch, physiological data, and contextual information, future systems may support user-specific calibration, intent prediction, and dynamic adjustment of feedback. This is particularly promising for underexplored applications beyond glove-based haptics, including near-body comfort objects, ambient textile interfaces, and full-body assistive garments.

Interdisciplinary advances may also support emerging directions such as underwater XR, affective communication, and cognitive or physiological augmentation (Liang et al. 2023; Hsu et al. 2023). However, these opportunities also raise challenges in power consumption, signal reliability, privacy, long-term comfort, explainability, and ethical use of physiological data. Future work should therefore pursue interdisciplinary collaboration not only to improve technical performance, but also to ensure that textile-XR systems remain usable, sustainable, and meaningful in real-world contexts.

6.4 Limitations

This study presents several limitations that should be considered when interpreting the findings. The review used a publication cutoff of September 2025, potentially overlooking emerging work or non-English sources, which may limit the comprehensiveness of trends (Tricco et al. 2018). The title-only search strategy improved precision but may have excluded relevant studies whose textile-XR contribution was not explicitly reflected in the title (Nunez et al. 2019).

User study limitations include the relatively small and XR-experienced sample and the reliance on concept illustrations, which influenced the envisioning of novel forms like curtains, as seen in lower ratings and feedback

(Jansen et al. 2022). Although the sensitivity analysis indicates that the sample provides adequate statistical resolution for detecting the targeted omnibus differences under the stated assumptions, the sample size remains insufficient for precise population-level estimation, subgroup analysis, or reliable detection of small pairwise differences among the 22 conditions.

Furthermore, participants evaluated visual concept representations and received only brief descriptions of the intended interactions rather than detailed technical explanations. Consequently, the ratings capture perceived affordances, expectations, familiarity, and existing mental models rather than actual technological feasibility, embodied comfort, or interaction performance. Visual framing may also have influenced how familiar or intrusive particular concepts appeared. Future studies should combine concept-level evaluation with technically informed explanations, functional prototypes, broader samples, and embodied evaluations.

7 Conclusion

This paper addressed three research questions concerning the role, design space, and user perception of smart textiles in XR.

For RQ1, the scoping review showed that current research is strongly centered on on-body textile systems, especially gloves, and is dominated by VR applications that emphasize haptic feedback, motion tracking, and gesture interaction. While this body of work demonstrates the value of textile-based embodied interaction, it also reveals comparatively limited exploration of AR/MR contexts and of near-body and off-body textile forms.

For RQ2, we translated the reviewed literature into a design space structured by Textile Form, Interaction Mode, and Application Domain. This framework contributes a transferable way to map existing work, identify clusters and absences, and support design ideation beyond dominant glove-based paradigms. The two design cases further illustrate how the framework can guide the development of both social and productivity-oriented XR applications.

For RQ3, the user study suggests that participants generally perceived familiar on-body forms and established interaction modes such as gesture, tactile, and mid-air interaction relatively favorably within the presented concept set, while more intimate or biodata-driven interactions raise concerns related to comfort, privacy, and practical integration. These findings point to the importance of grounding textile-XR innovation in user expectations and lived bodily experience rather than technical novelty alone. These findings should not be interpreted as direct validation of the corresponding technologies. Instead, they reveal how users' existing mental models may shape expectations toward emerging XR-textile interactions and highlight the need to distinguish perceived affordances from actual embodied experience.

Theoretically, this work contributes a structured view of smart textile XR interaction that bridges fragmented research strands. Methodologically, it demonstrates the value of combining evidence synthesis, framework building, and user-oriented reflection. Practically, it offers guidance for researchers and designers seeking to develop more embodied, acceptable, and context-sensitive XR experiences. Future work should further validate underexplored textile forms through functional prototypes, broader user populations, and longitudinal evaluations in real-world XR settings.

Appendix A

The corpus of smart textiles for XR research, see the link: https://docs.qq.com/sheet/DQWxMb1NmV1dHemJE?tab=000001

For the interview questionnaire example, see the link: https://form.typeform.com/to/aha4Kp8U

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Data availability The literature corpus used in this scoping review is provided in the Appendix A. The anonymized user-study data supporting the findings of this study are available from the corresponding author upon reasonable request.

Declarations

Competing interests The authors declare no competing interests.

Informed consent Informed consent was obtained from all individual participants included in the user study.

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References

Adams RJ, Ellington AL, Armstead K, Sheffield K, Patrie JT, Diamond PT (2019) Upper extremity function assessment using a glove orthosis and virtual reality system. OTJR Occup Particip Health 39(2):81–89

Adochiei I, Patlagica S, Voiculescu D, Stanculescu M, Drosu O, Vita V (2019) Virtual reality assisted hand rehabilitation through the use of a smart glove. In: 2019 E-health and bioengineering conference (EHB). IEEE, pp 1–4. https://doi.org/10.1109/EHB47216.2019.8969903

Alexandre R, Postolache O, Girão PS (2019) Physical rehabilitation based on smart wearable and virtual reality serious game. In: 2019 IEEE international instrumentation and measurement technology conference (I2MTC), pp 1–6. IEEE. https://doi.org/10.1109/I2MTC.2019.8826947

Ardaiz O, Baztan R, Dominguez U, Marzo A (2019) Facilitating physical contact in collocated virtual reality and real-world player games through wearable touchable interfaces. In: Extended abstracts of the annual symposium on computer–human interaction in play companion extended abstracts, pp 75–80. https://doi.org/10.1145/3341215.3356989

Bazelle A, Pourrier-Nunez H, Rignault M, Chang M (2018) Simulation of different materials texture in virtual reality through haptic gloves. In: SIGGRAPH Asia 2018 Posters, pp 1–2

Behnke L, Sanchez-Botero L, Johnson WR, Agrawala A, Kramer-Bottiglio R (2023) Dynamic hand proprioception via a wearable glove with fabric sensors. In: 2023 IEEE/RSJ international conference on intelligent robots and systems (IROS). IEEE, pp 149–154

Borja EF, Lara DA, Quevedo WX, Andaluz VH (2018) Haptic stimulation glove for fine motor rehabilitation in virtual reality environments. In: The 5th international conference on Augmented reality, virtual reality, and computer graphics, AVR 2018, Otranto, Italy, June 24–27, 2018, Proceedings, Part II 5. Springer, pp 211–229

Boutin J, Kamoonpuri J, Faieghi R, Chung J, Ribaupierre S, Eagleson R (2024) Smart haptic gloves for virtual reality surgery simulation: a pilot study on external ventricular drain training. Front Robot AI 10:1273631

Cai S, Ke P, Jiang S, Narumi T, Zhu K (2019) Demonstration of thermairglove: a pneumatic glove for material perception in virtual reality through thermal and force feedback. In: SIGGRAPH Asia 2019 emerging technologies, pp 11–12. ACM. https://doi.org/10.1145/3355049.3360529

Cannavò A, Pratticò FG, Ministeri G, Lamberti F (2018) A movement analysis system based on immersive virtual reality and wearable technology for sport training. In: Proceedings of the 4th international conference on virtual reality, pp 26–31. https://doi.org/10.1145/3198910.3198917
Chan J, Torah R (2022) E-textile haptic feedback gloves for virtual and augmented reality applications. Eng Proc 15(1):1
Chaudhari A, Lokesh R, Chheang V, Doshi SM, Barmaki RL, Cashaback JG, Thostenson ET (2024) Characterizing the sensing response of carbon nanocomposite-based wearable sensors on elbow joint using an end point robot and virtual reality. Sensors 24(15):4894
Chen K-Y, Lee J-D, Zhang T-J (2021) The body sensor suit with mixed reality interactive games. In: 2021 IEEE international conference on consumer electronics-Taiwan (ICCE-TW). IEEE, pp 1–2
Cheok AD (2002) Game-city: a ubiquitous large area multi-interface mixed reality game space for wearable computers...., 2002. (ISWC 2002).
Chong T, Shen I-C, Qian Y, Umetani N, Igarashi T (2021) Real-time image-based virtual try-on with measurement garment. In: SIGGRAPH Asia 2021 emerging technologies, pp 1–2. ACM. https://doi.org/10.1145/3476122.3484832
Chu NN, Yang C-M, Wu C-C (2012) Game interface using digital textile sensors, accelerometer and gyroscope. IEEE Trans Consum Electron 58(2):184–189
Civelek T, Fuhrmann A (2022) Virtual reality learning environment with haptic gloves. In: Proceedings of the 2022 3rd international conference on education development and studies, pp 32–36. https://doi.org/10.1145/3528137.3528142
Connelly L, Jia Y, Toro ML, Stoykov ME, Kenyon RV, Kamper DG (2010) A pneumatic glove and immersive virtual reality environment for hand rehabilitative training after stroke. IEEE Trans Neural Syst Rehabil Eng 18(5):551–559
Cortes D, Bermejo B, Juiz C (2024) The use of CNNs in VR/AR/MR/XR: a systematic literature review. Virtual Reality 28(3):154
Cui H, Kong D, Feng Z (2024) Eyeglove: enhancing smart glove with visual information to assist students in conducting chemistry experiments in mixed reality laboratory. In: International conference on computer animation and social agents. Springer, pp 329–341
Cui D, Mousas C (2021) Evaluating wearable tactile feedback patterns during a virtual reality fighting game. In: 2021 IEEE international symposium on mixed and augmented reality adjunct (ISMAR-Adjunct). IEEE, pp 328–333
Dai H, Chen Y, Wang M, Sun D, Lin H (2024) Design of multichannel tactile feedback digital glove for virtual reality. IEEE Sens Lett 8(3):1–4
Damian I, Obaid M, Kistler F, André E (2013) Augmented reality using a 3d motion capturing suit. In: Proceedings of the 4th augmented human international conference, pp 233–234. https://doi.org/10.1145/2459236.2459277
De Tinguy X, Pacchierotti C, Marchal M, Lécuyer A (2018) Enhancing the stiffness perception of tangible objects in mixed reality using wearable haptics. In: 2018 IEEE conference on virtual reality and 3D user interfaces (VR). IEEE, pp 81–90
Duarte Y, Rodriguez A (2025) Rethinking breath in VR: a performative approach to enhance user flow with bio-sensing wearable interfaces. In: Proceedings of the extended abstracts of the CHI conference on human factors in computing systems, pp 1–9
Fitzgerald D, Foody J, Kelly D, Ward T, Markham C, McDonald J, Caulfield B (2007) Development of a wearable motion capture suit and virtual reality biofeedback system for the instruction and analysis of sports rehabilitation exercises. In: 2007 29th annual international conference of the IEEE engineering in medicine and biology society, pp 4870–4874. IEEE. https://doi.org/10.1109/IEMBS.2007.4353431
Ghahremani Honarvar M, Latifi M (2017) Overview of wearable electronics and smart textiles. J Text Inst 108(4):631–652
Gießer C, Schmitt J, Schmücker V, Eiler TJ, Braun V, Brück R (2023) Conception of a native connection of data gloves with haptic feedback to an augmented reality-headset with a planned evaluation in a medical use case. Current Dir Biomed Eng 9:69–72. https://doi.org/10.1515/cdbme-2023-1018
Guneser MT, Aksakal SY, Coskunsu MA, Çakır KN, Şahin B (2025) Designing an interactive glove with a robotic arm for virtual reality education applications. In: 2025 9th international artificial intelligence and data processing symposium (IDAP). IEEE, pp 1–7
Häkkilä J, Colley A, Roinesalo P, Lappalainen T, Rantala I, Väyrynen J (2017) Wearable augmented reality display for wellness. In: Proceedings of the 6th ACM international symposium on pervasive displays, pp 1–2. https://doi.org/10.1145/3078810.3084348
Häkkilä J, Colley A, Roinesalo P, Väyrynen J (2017) Clothing integrated augmented reality markers. In: Proceedings of the 16th international conference on mobile and ubiquitous multimedia, pp 113–121. https://doi.org/10.1145/3152832.3152850
Hamari J (2021) Towards the next generation of extended reality wearables. In: ACM SIGCHI annual conference on human factors in computing systems. ACM, pp 1–7. https://doi.org/10.1145/3411763.3451697
Hinchet R, Vechev V, Shea H, Hilliges O (2018) Dextres: wearable haptic feedback for grasping in VR via a thin form-factor electrostatic brake. In: Proceedings of the 31st annual ACM symposium on user interface software and technology, pp 901–912. https://doi.org/10.1145/3242587.3242657
Hoang TN, Thomas BH (2012) Distance-based modeling and manipulation techniques using ultrasonic gloves. In: 2012 IEEE international symposium on mixed and augmented reality (ISMAR). IEEE, pp 287–288

Houzangbe S, Christmann O, Gorisse G, Richir S (2018) Integrability and reliability of smart wearables in virtual reality experiences: a subjective review. In: Proceedings of the virtual reality international conference-Laval virtual, pp 1–6. https://doi.org/10.1145/3234253.3234305
Hsu F-S, Wang T-M, Chen L-H (2023) Robust vision-based glove pose estimation for both hands in virtual reality. Virtual Reality 27(4):3133–3148
Hu Y, Wang S, Feng L, Liu B, Xiang Y, Li C, Wang D (2024) Design and implementation of the high-integrated smart gloves for virtual reality interaction. Sens Rev 44(6):629–637
Huang Y, Morris A, Brunet C (2024) Designing a multi-modal wearable mixed reality system for body awareness in workspaces. In: 2024 IEEE 4th international conference on human–machine systems (ICHMS). IEEE, pp 1–6
Islam GN, Ali A, Collie S (2020) Textile sensors for wearable applications: a comprehensive review. Cellulose 27(11):6103–6131
Jamieson AR, Singh I, Nguyen DV, Waghmare KC, Singh BG, Gu Y, Wijesundara MB (2023) Integrating an assistive soft robotic glove with an immersive virtual reality hand rehabilitation game. In: 2023 IEEE 11th international conference on serious games and applications for health (SeGAH), pp 1–8. IEEE. https://doi.org/10.1109/SeGAH57547.2023.10253774
Jansen P, Colley M, Rukzio E (2022) A design space for human sensor and actuator focused in-vehicle interaction based on a systematic literature review. Proc ACM Interact Mobile Wearable and Ubiquitous Technol 6(2):1–51
Jiang M, Wang Y, Nanjappan V, Bai Z, Liang H-N (2024) E-textiles for emotion interaction: a scoping review of trends and opportunities. Pers Ubiquit Comput 28(3):549–577
Jung K, Kim S, Oh S, Yoon SH (2024) Hapmotion: motion-to-tactile framework with wearable haptic devices for immersive VR performance experience. Virtual Reality 28(1):13
Jung S, Zhou M, Ma J, Yang R, Cramer SC, Dobkin BH, Yang LF, Rosen J (2024) Wearable body sensors integrated into a virtual reality environment-a modality for automating the rehabilitation of the motor control system. In: 2024 46th annual international conference of the IEEE engineering in medicine and biology society (EMBC), pp 1–4. IEEE
Kačerová I, Kubr J, Hořejší P, Kleinová J (2022) Ergonomic design of a workplace using virtual reality and a motion capture suit. Appl Sci 12(4):2150
Kammerlander RK, Pereira A, Alexanderson S (2021) Using virtual reality to support acting in motion capture with differently scaled characters. In: 2021 IEEE virtual reality and 3D user interfaces (VR). IEEE, pp 402–410
Kang D, Lee C-G, Kwon O (2023) Pneumatic and acoustic suit: multimodal haptic suit for enhanced virtual reality simulation. Virtual Reality 27(3):1647–1669
Karunanayaka K, Nijholt A, Halloluwa T, Ranasinghe N, Wickramasinghe M, Vyas D (2021) Multisensory augmented reality. In: IFIP conference on human-computer interaction. Springer, pp 558–563
Kim S, Gu S, Kim J (2022) Variable shape and stiffness feedback system for VR gloves using SMA textile actuator. Fibers Polym 23(3):836–842
Kim J, Lee S, Bae J (2025) An augmented reality-based wearable system for handheld-free and intuitive robot programming. J Intell Manuf 37(3):1023–1032
Kirby J, Kim K (2025) Design and evaluation of haptic glove-based text inputs in virtual reality. In: 2025 25th international conference on digital signal processing (DSP). IEEE, pp 1–5
Kishishita Y, Das S, Ramirez AV, Thakur C, Tadayon R, Kurita Y (2019) Muscleblazer: Force-feedback suit for immersive experience. In: 2019 IEEE conference on virtual reality and 3D user interfaces (VR). IEEE, pp 1813–1818
Kiuchi A, Qi A, Li EM, Maruscsak D, Sandor C, Igarashi T (2023) Perfectfit: custom-fit garment design in augmented reality. In: SIGGRAPH Asia 2023 XR, pp 1–2. ACM. https://doi.org/10.1145/3610549.3614592
Ko SH, Rogers J (2021) Functional materials and devices for XR (VR/AR/MR) applications
Ku J, Mraz R, Baker N, Zakzanis KK, Lee JH, Kim IY, Kim SI, Graham SJ (2003) A data glove with tactile feedback for FMRI of virtual reality experiments. Cyberpsychol Behav 6(5):497–508
Kuo F-L, Lee H-C, Kuo T-Y, Wu Y-S, Lee Y-S, Lin J-C, Huang S-W (2023) Effects of a wearable sensor-based virtual reality game on upper-extremity function in patients with stroke. Clin Biomech 104:105944
Kuusk K, Langereis G, Tomico O (2013) Bedtime stories: weaving traditions into digital technologies. Nordes 1(5)
Kwon D, Choi H, Cho H, Lee J, Kim G (2019) Pillowvr: virtual reality in bed. Proceedings of the 25th ACM symposium on virtual reality software and technology, pp 1–2. https://doi.org/10.1145/3359996.3365029
Lee KT, Chee PS, Lim EH, Kam YH (2023) Data glove with integrated polyethylene-carbon composite-based strain sensor for virtual reality applications. Chem Eng Technol 46(12):2480–2486
Lee S, Park K, Lee J, Kim K (2017) User study of VR basic controller and data glove as hand gesture inputs in VR games. In: 2017 international symposium on ubiquitous virtual reality (ISUVR). IEEE, pp 1–3
Li F, Chen J, Ye G, Dong S, Gao Z, Zhou Y (2023) Soft robotic glove with sensing and force feedback for rehabilitation in virtual reality. Biomimetics 8(1):83
Liang Q, Zhang D, He T, Zhang Z, Wang H, Chen S, Lee C (2023) Fiber-based noncontact sensor with stretchability for underwater wearable sensing and VR applications. ACS Nano 18(1):600–611
Libanori A, Chen G, Zhao X, Zhou Y, Chen J (2022) Smart textiles for personalized healthcare. Nat Electron 5(3):142–156

Li R, Lin W, Huo Z, Ren G, Cheng Z, Wang G, Dong S, Li J (2023) Investigating safety and efficiency of wearable mixed reality helmet interaction design for delivery riders in multitasking environment. In: 2023 IEEE 5th Eurasia conference on biomedical engineering, healthcare and sustainability (ECBIOS), pp 83–87. IEEE
Li R, Liu Y, Qian X (2025) Toward wearable cybersickness detection: an EEG-based classification framework leveraging commercial VR games. In: 2025 18th international congress on image and signal processing, BioMedical engineering and informatics (CISP-BMEI), pp 1–6. IEEE
Lin C-Z, Cheng J-H (2025) Integrating the pdca cycle assessment method into the development of MR pneumatic glove rehabilitation aids: a case study in hand rehabilitation for stroke patients. In: International conference on human-computer interaction. Springer, pp 257–271
Lin C-J, Liu R-S (2025) Virtual reality with haptic gloves for human–robot collaborative assembly. In: 2025 IEEE international conference on advanced robotics and its social impacts (ARSO). IEEE, pp 355–360
Lin J, She Y, Lin L, Chen S, Chen J, Liu H, Xu X, Shen Y, Li J (2022) Out of theater: Interactive mixed-reality performance for intangible culture heritage glove puppetry. In: Proceedings of the 10th international symposium of Chinese CHI, pp 181–189. https://doi.org/10.1145/3565698.3565782
Liu S, Ma K, Yang B, Li H, Tao X (2021) Textile electronics for VR/AR applications. Adv Func Mater 31(39):2007254
Liu M, Wilder S, Sanford S, Dewil S, Harel N, Nataraj R (2023) Multimodal augmented feedback for functional grasp training using a smart glove and virtual reality for persons with spinal cord injury. In: 2023 9th international conference on virtual reality (ICVR). IEEE, pp 435–440. https://doi.org/10.1109/ICVR57957.2023.10169674
Liu H, Zhang Z, Xie X, Zhu Y, Liu Y, Wang Y, Zhu S-C (2019) High-fidelity grasping in virtual reality using a glove-based system. In: 2019 international conference on robotics and automation (ICRA), pp 5180–5186. IEEE. https://doi.org/10.1109/ICRA.2019.8794230
M Haikal MFHB, Ghosh DK, Lay JK, Caluya NR, Chandler DM (2025) Chordfusion: interactive piano training in mixed reality with custom gloves. In: International conference on entertainment computing, pp 478–486. Springer
Mangili P, Netti V (2024) Design and development of a space suit mock-up for VR-based EVA research and simulation. J Space Saf Eng 11(4):622–627
Mann S (1997) Wearable computing: a first step toward personal imaging. Computer 30(2):25–32
Masmoudi M, Zenati N, Benbelkacem S, Hadjadj Z, Djekoune O, Guerroudji MA, Izountar Y (2021) Low-cost haptic glove for grasp precision improvement in virtual reality-based post-stroke hand rehabilitation. In: 2021 international conference on artificial intelligence for cyber security systems and privacy (ai-csp). IEEE, pp 1–3
Meng X, Feng Z, Tian J (2020) A smart glove of combining virtual and real environments for chemical experiment. In: Proceedings of 2020 6th international conference on computing and data engineering, pp 227–231. https://doi.org/10.1145/3379247.3379263
Muthukumarana S, Nassani A, Park N, Steimle J, Billinghurst M, Nanayakkara S (2022) Xrtic: a prototyping toolkit for XR applications using cloth deformation. In: 2022 IEEE international symposium on mixed and augmented reality (ISMAR), pp 548–557. IEEE
Najm A, Banakou D, Michael-Grigoriou D (2024) Development of a modular adjustable wearable haptic device for XR applications. In: Virtual Worlds, vol 3. MDPI, pp 436–458
Nakao K, Morimoto N (2025) Investigating 3d object identifiability in VR using a glove-type vibrotactile feedback system. In: 2025 IEEE international conference on consumer electronics-Taiwan (ICCE-Taiwan). IEEE, pp 821–822
Nanjappan V, Shi R, Liang H-N, Xiao H, Lau KK-T, Hasan K (2019) Design of interactions for handheld augmented reality devices using wearable smart textiles: findings from a user elicitation study. Appl Sci 9(15):3177
Narejo A, Baqai A, Sikandar N, Absar A, Narejo S (2020) Physiotherapy: Design and implementation of a wearable sleeve using IMU sensor and VR to measure elbow range of motion. Int J Adv Comput Sci Appl **11**(9)
Nunez E, Hirokawa M, Perusquia-Hernandez M, Suzuki K (2019) Effect on social connectedness and stress levels by using a huggable interface in remote communication. In: 2019 8th international conference on affective computing and intelligent interaction (ACII), pp 1–7. IEEE
Oota S, Murai A, Mochimaru M (2019) Lucid virtual/augmented reality (lvar) integrated with an endoskeletal robot suit: Stillsuit: A new framework for cognitive and physical interventions to support the ageing society. In: 2019 IEEE conference on virtual reality and 3D user interfaces (VR). IEEE, pp 1556–1559
Piekarski W, Smith R (2006) Robust gloves for 3d interaction in mobile outdoor ar environments. In: IEEE/ACM international symposium on mixed and augmented reality, pp 251–252. https://doi.org/10.1109/ISMAR.2006.297830IEEE
Pontreau M, Louison C, Oréfice P-H, Bouchigny S, Le T-LS, Gueorguiev D, Panëels, S (2025) Demonstrating wavy: a hand wearable with force and vibrotactile feedback for multimodal interaction in virtual reality. In: 2025 IEEE conference on virtual reality and 3D user interfaces abstracts and workshops (VRW), pp 1652–1653. IEEE
Priebe M, Foo E, Holschuh B (2020) Shape memory alloy haptic compression garment for media augmentation in virtual reality environment. In: Adjunct Proceedings of the 33rd annual ACM symposium on user interface software and technology, pp 34–36. https://doi.org/10.1145/3379350.3416177

Qian J, Young-Ng M, Li X, Cheung A, Yang F, Huang J (2020) Portalware: A smartphone-wearable dual-display system for expanding the free-hand interaction region in augmented reality. In: Extended abstracts of the 2020 CHI conference on human factors in computing systems, pp 1–8. https://doi.org/10.1145/3334480.3383079
Qiu Y, He X, Li Z, Peng Z, Huang Y, Yu X (2024) Wearable sensors for motion and electrophysiological signal tracking in XR. Korean J Chem Eng 42(9):1951–1976
Quaresma C, Gomes M, Cardoso H, Ferreira N, Vigário R, Quintão C, Fonseca M (2019) An integrated system combining virtual reality with a glove with biosensors for neuropathic pain: a concept validation. In: Advances in human factors and systems interaction: proceedings of the AHFE 2018 international conference on human factors and systems interaction, July 21-25, 2018, Loews Sapphire Falls Resort at Universal Studios, Orlando, Florida, USA 9. Springer, pp 274–284
Rao AK, Choudhary G, Negi R, Dutt V (2023) Is virtual reality better than desktop-based cognitive training? A neurobehavioral evaluation of visual processing and transfer performance. In: 2023 IEEE international symposium on mixed and augmented reality adjunct (ISMAR-Adjunct). IEEE, pp 308–314
Rauschnabel PA, Felix R, Hinsch C, Shahab H, Alt F (2022) What is XR? towards a framework for augmented and virtual reality. Comput Hum Behav 133:107289
Ruckdashel RR, Venkataraman D, Park JH (2021) Smart textiles: a toolkit to fashion the future. J Appl Phys 129(13):130903
Rumiński D, Klinker G (2018) Augmenting mixed reality applications with the vibro motors wearable. In: IEEE international symposium on mixed and augmented reality adjunct (ISMAR-Adjunct). IEEE, pp 401–402
Sakata K, Shizuki B, Kawaguchi I, Takahashi S (2021) A hat-shaped pressure-sensitive multi-touch interface for virtual reality. In: Proceedings of the 27th ACM symposium on virtual reality software and technology, pp 1–3. https://doi.org/10.1145/3489849.3489909
Saladtook S, Rujirotthamrong P, Eiwaroon C, Phunpar J, Saladtook C, Kerdvibulvech C (2025) Haptic gamer suit for enhancing VR games experience. Augment Hum Res 10(1):6
Samala AD, Bojic L, Soha R, Miftachul Arif Y, Tsoy D, Pereira Coelho D (2024) Extended reality for education: mapping current trends, challenges, and applications. Jurnal Pendidikan Teknologi Kejuruan 7(3):140–169
Sarkar D, Kumar A, Arora A (2025) A soft glove with proprioceptive sensing and multi-modal haptic feedback for VR and telerobotic applications. J Bionic Eng 22(2):684–702
Sasing VJA, See AR, Tiong TJT, Alcuitas ABR, Kuo H-H, Seepold, R (2025) Development of cutaneous feedback haptic glove for vr industrial training. In: 2025 1st international conference on consumer technology (ICCT-Pacific), pp 1–4. IEEE
Seo S-W, Jung W-S, Kim Y (2024) 3d hand motion generation for VR interactions using a haptic data glove. Multimodal Technol Interact 8(7):62
Shi J, Liu S, Zhang L, Yang B, Shu L, Yang Y, Ren M, Wang Y, Chen J, Chen W (2020) Smart textile-integrated microelectronic systems for wearable applications. Adv Mater 32(5):1901958
Shor D, Zaaijer B, Ahsmann L, Immerzeel S, Weetzel M, Eikelenboom D, Hartcher-O'Brien J, Aschenbrenner D (2018) Designing haptics: comparing two virtual reality gloves with respect to realism, performance and comfort. In: 2018 IEEE international symposium on mixed and augmented reality adjunct (ISMAR-Adjunct). IEEE, pp 318–323
Silva L, Dantas R, Pantoja A, Pereira A (2013) Development of a low cost dataglove based on arduino for virtual reality applications. In: 2013 IEEE international conference on computational intelligence and virtual environments for measurement systems and applications (CIVEMSA). IEEE, pp 55–59
Simon TM, Smith RT, Thomas BH (2014) Wearable jamming mitten for virtual environment haptics. In: Proceedings of the 2014 ACM international symposium on wearable computers, pp 67–70. https://doi.org/10.1145/2634317.2634342
Sousa M, Vieira J, Medeiros D, Arsenio A, Jorge J (2016) Sleevear: augmented reality for rehabilitation using realtime feedback. In: Proceedings of the 21st international conference on intelligent user interfaces, pp 175–185. https://doi.org/10.1145/2856767.2856773
Stanney KM, Nye H, Haddad S, Hale KS, Padron CK, Cohn JV (2021) Extended reality (XR) environments. In: Handbook of human factors and ergonomics, pp 782–815
Starner T (2002) The challenges of wearable computing: part 2. IEEE Micro 21(4):54–67
Strachan N, Kalyanji J, Lowe A, Barde A, Billinghurst M (2023) Fabric electrodes for physiological sensing in a VR HMD. In: Proceedings of the 29th ACM symposium on virtual reality software and technology, pp 1–2. https://doi.org/10.1145/3611659.3616888
Ta K, Sharlin E, Oehlberg L (2018) Bod-ide: an augmented reality sandbox for efashion garments. In: Proceedings of the 2018 ACM conference companion publication on designing interactive systems, pp 33–37. https://doi.org/10.1145/3197391.3205408
Thilakarathna R, Phlernjai M (2025) Finger tracking for wearable VR glove using flexible rack mechanism. Virtual Reality Intell Hardw 7(1):1–25
Thomas BH, Smith M, Simon T, Park J., Park J, Von Itzstein GS, Smith RT (2011) Glove-based sensor support for dynamic tangible buttons in spatial augmented reality design environments. In: 2011 15th annual international symposium on wearable computers, pp 109–110. IEEE. https://doi.org/10.1109/ISWC.2011.25

Tomico O, Wilde D (2016) Soft, embodied, situated & connected: enriching interactions with soft wearables. mUX J Mobile User Exp 5(1):3
Tran TTM, Brown S, Weidlich O, Billinghurst M, Parker C (2023) Wearable augmented reality: Research trends and future directions from three major venues. IEEE Trans Visual Comput Graph 29(11):4782–4793
Tricco AC, Lillie E, Zarin W, O'Brien KK, Colquhoun H, Levac D, Moher D, Peters MD, Horsley T, Weeks L (2018) Prisma extension for scoping reviews (PRISMA-ScR): checklist and explanation. Ann Intern Med 169(7):467–473
Uğur S (2013) Wearing embodied emotions: a practice based design research on wearable technology. Springer, Berlin
Vanichvoranun N, Lee H, Kim S, Yoon SH (2024) Estatig: wearable haptic feedback with multi-phalanx electrostatic brake for enhanced object perception in VR. Proc ACM Interact Mobile Wearable Ubiquitous Technol 8(3):1–29
Vasarainen M, Paavola S, Vetoshkina L (2021) A systematic literature review on extended reality: virtual, augmented and mixed reality in working life. Int J Virtual Reality 21(2):1–28
Villarreal-Narvaez S, Septon T, Dumas B (2025) Designing touch-based gestures for thigh-worn wearable device in passenger-focused vehicular XR environments. In: Companion Proceedings of the 17th ACM SIGCHI Symposium on Engineering Interactive Computing Systems, pp 33–41
Wang K, Yap LW, Gong S, Wang R, Wang SJ, Cheng W (2021) Nanowire-based soft wearable human-machine interfaces for future virtual and augmented reality applications. Adv Func Mater 31(39):2008347
Wang J, Suo J, Song Z, Li WJ, Wang Z (2023) Nanomaterial-based flexible sensors for metaverse and virtual reality applications. Int J Extreme Manuf 5(3):032013
Wang H, Feng Z, Meng X (2023) Multimodal information perception and understanding: application of smart glove in virtual-reality fusion chemistry experiment platform. Adv Mater Technol 8(10):2200549
Way D-L, Lau W-K, Huang TY (2019) Glove puppetry cloud theater through a virtual reality network. In: ACM SIGGRAPH 2019 Posters, pp 1–2. ACM. https://doi.org/10.1145/3306214.3338564
Wen F, Sun Z, He T, Shi Q, Zhu M, Zhang Z, Li L, Zhang T, Lee C (2020) Machine learning glove using self-powered conductive superhydrophobic triboelectric textile for gesture recognition in vr/ar applications. Adv Sci 7(14):2000261
Weng W, Chen P, He S, Sun X, Peng H (2016) Smart electronic textiles. Angew Chem Int Ed 55(21):6140–6169
Xia Y, Li W, Duan S, Lei W, Wu J (2022) Low-cost, light-weight scalable soft data glove for VR applications. In: 2022 5th international conference on circuits, systems and simulation (ICCSS). IEEE, pp 202–205
Yang L, Jiang J, Wang R, Mao Z, Fang L, Qi Y, Zhang J, Wu C, Yu H (2023) Fully flexible smart gloves and deep learning motion intention prediction for ultra-low latency VR interactions. IEEE Sens Lett 7(9):1–4
Yessentayev T, Mukanova Z (2024) Consumer grade VR gloves based on hall effect sensors in an angular flexion arrangement. In: 2024 IEEE AITU: digital generation. IEEE, pp 98–103
Yoon SH, Sung Y, Song KW, Jung K, Seo K, Kim J, Yi HI, Vanichvoranun N, Jeong H, Lee H (2024) Adaptive and immersive XR interactions with wearable interfaces (demo of Kaist Hci tech lab). In: Extended abstracts of the CHI conference on human factors in computing systems, pp 1–4
Yu F, Yu C, Tian Z, Liu X, Cao J, Liu L, Du C, Jiang M (2024) Intelligent wearable system with motion and emotion recognition based on digital twin technology. IEEE Internet Things J 11(15):26314–26328
Zhang Z, He T, Zhu M, Sun Z, Shi Q, Zhu J, Dong B, Yuce MR, Lee C (2020) Deep learning-enabled triboelectric smart socks for IoT-based gait analysis and VR applications. npj Flexible Electron 4(1):29
Zhu M, Sun Z, Zhang Z, Shi Q, He T, Liu H, Chen T, Lee C (2020) Haptic-feedback smart glove as a creative human-machine interface (HMI) for virtual/augmented reality applications. Sci Adv 6(19):8693
Zou R, Liu Y, Zhao J, Cai H (2024) Multimodal learning-based proactive human handover intention prediction using wearable data gloves and augmented reality. Adv Intell Syst 6(4):2300545
Zwicky F (1967) The morphological approach to discovery, invention, research and construction. In: New methods of thought and procedure: contributions to the symposium on methodologies. Springer, pp 273–297

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Authors and Affiliations

Mengqi Jiang[1] · Tianchen Dai[1] · Ruoqing Yan[1] · Rongkai Shi[2] · Ziqian Bai[3] · Qianwen Tan[3] · Jun Huang[4] · Hai-Ning Liang[2]

✉ Hai-Ning Liang
hainingliang@hkust-gz.edu.cn

Mengqi Jiang
jiangmengqi@hit.edu.cn

Tianchen Dai
daitianchen@hit.edu.cn

Ruoqing Yan
25S066012@stu.hit.edu.cn

Rongkai Shi
rongkaishi@hkust-gz.edu.cn

Ziqian Bai
baizq@sustech.edu.cn

Qianwen Tan
12332673@mail.sustech.edu.cn

Jun Huang
huang.jun16@u.nus.edu

[1] Future Design School, Harbin Institute of Technology (Shenzhen), Nanshan, Shenzhen 518000, Guangdong, China

[2] Computational Media and Arts Thrust, Hong Kong University of Science and Technology (Guangzhou), Nansha, Guangzhou 510000, Guangdong, China

[3] School of Automation and Intelligent Manufacturing, Southern University of Science and Technology, Nanshan, Shenzhen 518000, Guangdong, China

[4] Institute of Systems Science, National University of Singapore, 25 Heng Mui Keng Terrace, Singapore 510000, Singapore